Regimes' Characteristics and Time Series Forecasting

FGV-EESP Masters' Thesis

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This thesis investigates how regime switching models learn and forecast when the data-generating process is mis-specified, and whether observable regime characteristics can help explain and predict model performance. I introduce a unified framework separating the *series generating process* and the *regime generating process*, and formalize *regime-conditional metrics* (RC) that summarize regime distributions.

A Monte Carlo setup is defined to simulate series, estimate models, and calculate RC metrics. The framework and implementation is general and expandable, but I focus on stationary $AR(1)$ series, with two regimes governed by Markov Switching (MS), Self-Exciting Threshold (SET), or Smooth Transition (ST) mechanisms, and utilize the 1st, 2nd moments, and lag-1 autocorrelation for RC metrics.

The results show that RC metrics meaningfully characterize regime separation, with MS RGPs there is a clear map from the metrics values and which parameter is changing, but not for SET and ST. SET and ST generally yield better fixed-effect forecasting performance, while MS is more flexible and performs better in asymmetric regimes cases. Model mis-specification raises RMSE (baseline increase of 0.547, larger with volatility changes), but RGP interactions are insignificant. Given RC metrics, SET/ST perform best when average separation is high, MS suffers with high ACF separation. Correct regime identification typically narrows RMSE tails (though not uniformly).

The study's main limitation is external validity: findings are conditional on a restricted pool of DGPs. Expanding to richer DGPs, more regimes, different error distributions, and including more RC metrics is the natural next step. Nevertheless, the RC-metric approach offers a promising diagnostic and model-selection tool for applied econometricians working with regime switching time series.

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1 Introduction

Regime switching (RS) models describe time series that exhibit different behavior – different parameters – across different regimes. They are useful to capture non-linearities in time series, having been widely used in economics and finance, for instance, to model business cycles and market volatility. There are several types of regime switching modeling, some with stochastic switching, such as Markov-switching models, and some with deterministic switching, such as threshold models.

As with any forecasting model, it is important to understand the factors that influence their performance, and how econometricians can use this knowledge to improve their models. In this work, I focus on: (i) the ability of these models to learn and generate performant forecasts under the presence of mis-specification; and (ii) how that ability relates to the characteristics of the regimes.

The first focus is common in forecasting econometrics: exactly identifying the data generating process (DGP) is the exception, not the rule, so the modeling goal is actually to find a robust approximator. It is important to document how each RS model behaves under different mis-specifications, explore possible universal approximators, and understand how each element of the DGP affects the learning problem of the models.

The second focus is less orthodox and specific to RS models. These models are special in the sense that they not only identify the series in question but also its states – its regimes – thus allowing the econometrician to describe the distribution of each regime and how different they are from each other. This characterization of regimes' distributions might be informative for the model's performance: for example, if the DGP implies different intercepts across regimes, a model whose identified regimes have the same conditional average is probably not capturing that dynamic well; or some class of model can be good at capturing that dynamic but bad at capturing changes in persistence. These examples might seem obvious, but I will show that there is much useful information to be taken from this kind of analysis.

The nature of this project is explorative. I will simulate a diverse set of DGPs and try to find stylized facts about how each RS model adjusts to them, and how the characteristics of the estimated regimes relate to this adjustment. To make things more concrete, in the remainder of this section I synthesize the methodology, describe the patterns I hope to find, and present some of the actual findings.

1.1 Basic methodology and hypothesis

The methodology follows a common setup. The first step is to establish a theoretical framework that describes all RS models in a unified way. Here, I denote the separate 'ingredients' in an RS DGP: the *series generating process* (SGP), the *regime generating process* (RGP), and what changes across regimes, the *regimes nature* (RN). By varying these 'ingredients', one can define a diverse set of DGPs to be studied. I define the notion of regime conditional (RC) metrics and the different ways that they can be calculated and compared. I propose a Monte Carlo setup to generate series, estimate models, and calculate RC metrics. The code is available and implemented in a similarly modular and expandable way.

Creating a general framework was a goal in itself, but for this work, I considered only a specific set of DGPs, models, and metrics, answering only some of the questions it allows to be asked.

I focus on stationary $AR(1)$ processes, with regime switching via Markov Switching, Self-Exciting Threshold, and Smooth Transition, with symmetric and asymmetric variations. There, the regimes can differ in one of the three parameters of the $AR(1)$, and the change can be 'big' or 'small'. A no-RS model is included as a baseline.

For the analysis, the first step is exploratory, to understand how the regimes' distributions differ across these DGPs. Do the metrics correctly identify the differences between the distributions? Can they help us get stylized facts to better understand how these models work?

Then, the focus is to understand if the distribution of regimes informs something about the models' performance. That is, do the models fare better when faced with a specific profile of regimes' distributions? Or does matching the distributions improve models' performance in the sense of the metrics?

For the more orthodox objectives, we first explore the general performance of the considered models in this specific pool of DGPs. Which one performs better? How do the models' components relate to that performance, and how does it vary across DGP scenarios (e.g. asymmetric DGPs)? Does mis-specifying the DGP impact performance, and how? Which model component (fit, regimes, parameters, etc.) is more important to match, in terms of performance?

The goal of this work is to answer these questions, generating stylized facts about the DGPs, and practical recommendations for modeling regime switching time series. Results are conditional in the population of DGPs considered, but are still useful. Notably: the interaction between RGPs and RNs is paramout to the behaviour of the series, with MS RGPs it can be profiled by the dispersion of the RC metrics, but not with SET and ST; MS models are more flexible and deal better with scenarios such as asymmetric RGPs, but SET and ST are otherwise overall better; Mis-specifying the RGP increases the RMSE by 0.547, and more with changes in volatility, but RGPs interactions are insignificant.

An initial analysis of how a model fares for each profile of RC metrics is done, with potential for powerful practical utility for model selection, e.g. SET and ST are best on series with high average separation, MS does worse with high ACF separation, while the opposite is true for ST. But, this analysis has issues of external validity and variance across non-observable DGP factors, and requires further investigation.

The rest of this work is divided as follows: Section 2 presents the literature review. The general framework is presented in Section 3 and Section 4, while the specific implementation chosen is presented in Section 5. The results are split into a more exploratory section (6) and a systematic one (7). Finally, Section 8 concludes.

2 Related literature

The regime switching literature is vast and contains many model variations. It is important to map the models, the similarities, and the differences between them. Additionally, an important starting point is to discuss what is already known about their forecasting performance and the factors that influence it. Each is done in the sections below. Before doing so, I will better define the bounds of RS literature by discussing two closely related ones.

The first is the state-space (SS) literature, with its quintessential implementation by Kalman (1960). While RS and SS models have developed as somewhat independent fields, RS can be viewed as a subset of SS, where the state (regime) variable and the observed series variable are modeled separately. This separation is central to the framework used in this paper. Bridges between the literatures include Switching State-Space Models and the seminal work by Kim (1994), which extends Hamilton's Markov-switching model to general state-space models.

The second is the structural break (SB) literature, the most relevant starting point being Chow (1960). Much of it is devoted to diagnosing breaks, which are indeed present in RS settings, with the non-constant parameters. However, SB models typically treat breaks as exogenous and non-recurring. Bridging the gap, Bai; Perron (1998) allows for multiple unknown breaks, which can be relevant for RS contexts, while Chib (1998) demonstrates that SBs can be formulated as Markov-switching processes that have only positive probability for staying in the initial regime and switching to the next, not for switching back. Ultimately, I found that SB models were not comparable with RS models for the analysis done in this work.

2.1 Regime switching models

Two of the most essential aspects of the different RS approaches are: (i) if the latent regime variable is modeled in a deterministic or stochastic fashion, and (ii) if the changes between regimes are abrupt or smooth.

The most common deterministic models are the threshold-based ones, where some observable variable being above or below some threshold(s) is what determines the regime. The work of Tong (1978) and (1980) popularized the threshold autoregressive model, each regime having its own set of autoregressive parameters. Tong proposed that capturing smooth transitions between regimes would be important, and Terasvirta; Anderson (1992), (1994) defined the smooth transition autoregressive model, where the distance between an observable variable and some threshold determines the continuous weight of each regime.

On the stochastic front, the Markov switching literature started with Hamilton (1989) via the MSAR model, where the regime is governed by an unobservable Markov process – the probability of switching to another regime is constant and depends only on the current regime. This implies a geometric distribution for the number of periods in a given regime's instance.[1] The Markov switching smooth transition model, as defined by Elliott; Siu; Lau (2018), exists, but has more added complexity than the very natural jump from the TAR to the STAR model.

2.1.1 Variations of the classic models

Moving forward, many variations on the regime variable modeling were created. The threshold variable can have a delay or some transformation; it can be the series itself, an exogenous variable, or even a

[1] Throughout this document, 'regime instance' will be used to describe a contiguous period of time without switches. In a given series, a given regime can have several instances.

non-linear combination of variables (Chen; Liu; So, 2011). The probability distributions of MS models were extended to allow different distributions for the time spent in one regime, and dependence on more past values (Ferguson, 1980). The smooth transition function has several options, with common ones being the logistic and the exponential. The models were generalized to any number of regimes, with the STAR model being equivalent to a Neural Network where the S regimes are S nodes in the hidden layer, as described by Medeiros (2000).

Blurring the line between deterministic and stochastic models, Chang; Choi; Park (2017) uses threshold dynamics but adds an innovation that is dependent on the previous state's innovation, and simplifies to a MS model when the threshold dynamic is exogenous and stationary. Wu; Chen (2007) creates a half-threshold, half-random regime process. There are also unsupervised approaches to estimation, that make no assumption on the nature of the latent process, as by Akioyamen; Tang; Hussien (2020), where some clustering model can be used to identify regimes, and later the functional form can be estimated for each regime separately.

As I will note in this work, the functional form across regimes and the regime process itself are fairly independent, thus other variations arise from considering more complex functions than the autoregressive one. ARMA models have their RS counterparts (Brockwell; Liu; Tweedie, 1992). Not only the mean, but the variance can also be modeled: the ARCH/GARCH family, very relevant for finance, have their regime switching versions (Chen; Liu; So (2011)). More recently, models such as decision trees have been adapted to the regime switching context, as by Adam; Ötting; Michels (2024). Similarly, there are also models for vectors of time series.

General reviews on RS models include (Tan; Wu, 2025), (Potter, 2000), and (Hamilton, 2020), while (Chen; Liu; So, 2011) focuses on threshold models, (Dijk; Teräsvirta; Franses, 2002) in smooth transition, and Song; Woźniak (2021) in Markov switching. Note that RS models are considered in both frequentist and Bayesian frameworks, with the latter inheriting a lot from the SS models estimation literature.

2.2 Forecasting performance

There are many research topics in RS performance. I'll focus on (i) important factors that relate to model selection and hyperparametrization, to contextualize the decisions I made in this work; and (ii) comparisons between models, to contextualize the experiments I ran.

2.2.1 Hyperparametrization

While RS models are frequently cited for their superior in-sample fit, which is useful for explaining historical phenomena, Dacco; Satchell (1999) noted that even minor errors in forecasting the future regime state can propagate through the non-linear structure, causing the overall prediction to perform worse than linear alternatives. Furthermore, standard metrics like mean squared error may be ill-suited for evaluating non-linear time series, potentially masking the utility of these models in capturing turning points or specific economic states.

A primary challenge in RS modeling is managing the trade-off between flexibility and overfitting. The most critical decision is on the number of regimes: too few can underfit, while too many might lead to overparameterization. Similarly, allowing all parameters to switch can help capture complex dynamics and avoid mis-specification, but doing it when unneeded often dilutes out-of-sample power (Tan; Wu, 2025).

Each model also has its own specificities. For Markov Switching models, the estimation method is relevant: for example, EM algorithms have been noted to balance accuracy and speed in high-dimensional settings (Akbal, 2024). Moreover, the translation of soft posterior probabilities into hard regime labels affects accuracy, and different rules have different properties (Hall; Pagan, 2025). For deterministic models, the challenge lies in variable selection: identifying the correct threshold variable, delay parameter, or non-linear combination of variables remains a significant hurdle for effective specification.

2.2.2 Comparisons between models

Many papers compare the different RS models in many different contexts (Clements; Krolzig, 1998), (Bierbrauer; Trück; Weron, 2004), (Pinson et al., 2008), (Janczura; Weron, 2010), (Elias; Wahab; Fang, 2014), (Chen; Bunn, 2014), (Panopoulou; Pantelidis, 2015), (Verne, 2021), (Aydın; Mermi, 2022). No single model is universally superior, the same context can present a different 'best' model depending on the focus (e.g. nowcasting, regime identification, portfolio performance, etc.) (Akbal, 2024).

TAR models are best employed when regime changes are triggered by a single, observable variable with rigid boundaries. They've shown effective for financial assets like gold prices and exchange rates, where transitions are fast rather than gradual (Aydın; Mermi, 2022). However, their reliance on observable triggers is a limitation: in contexts like offshore wind power, where fluctuations are driven by complex, non-observable states, TAR models fail to capture the underlying dynamics and significantly underperform compared to latent variable models (Pinson et al., 2008).

STAR models are theoretically appropriate for gradual economic adjustments but often face practical identification challenges. In many financial applications, the estimated smoothness parameter becomes so high that the model collapses into an abrupt threshold model, rendering the specific 'smooth' specification inefficient (Aydın; Mermi, 2022). However, STAR models can outperform MS in macroeconomic contexts characterized by explosive volatility, such as GDP growth requiring the capture of 'brutal' transitions typical of recession phases (Verne, 2021).

Markov-Switching Models (MS/MSAR) are the superior choice when regimes are driven by latent, multi-factor variables (e.g., market sentiment or meteorology) rather than a single observable index. But, with a more flexible regime framework, they have shown to be more sensitive to specification of the number of regimes Janczura; Weron (2010).

3 Theoretical framework

In this section, I define the theoretical framework that guides the rest of this work. First, I define the general structure of RS DGPs, aligning all in a common mathematical representation, and relate the concepts of models and metrics to it. An important idea is the separation of the DGP into RGP and SGP.

3.1 The general regime switching DGP

Let $y_t \in \mathbb{R}$ denote the series of interest at time $t \in 1 : T$,[2] $T \in \mathbb{N}$. Let $S \in \mathbb{N}$ denote the number of regimes. The *regime variable* is a vector $r_t \in \mathbb{R}^S$ of 'weights' for each regime, indexed by r_t^s, $s \in 1 : S$.

In this work, I consider only univariate series.

A DGP can be written in terms of a pair: *regime generating process* (RGP) and *series generating process* (SGP). This is essentially the separation between the state/system equation, and the output/measurement equation, in state space models. They are functions with parameters Θ_r and Θ_y, respectively, such that:

$$
\begin{aligned}
r_t &= \mathrm{rgp}(\; y_{1:(t-1)}, \; r_{t-1}, & t \;;\; \Theta_r \;) \\
y_t &= \mathrm{sgp}(\; y_{1:(t-1)}, \; r_t, & t \;;\; \Theta_y \;) \\
&= \mathrm{sgp}(\; y_{1:(t-1)}, \; \mathrm{rgp}(y_{1:(t-1)}, r_{t-1}, t; \Theta_r), & t \;;\; \Theta_y \;)
\end{aligned}
\tag{3.1}
$$

Without loss of generality, I restrict the regime weights to be non-negative and sum to one, i.e. $r_t \in [0,1]^S$ and $\sum_{s=1}^{S} r_t^s = 1$. Notably, the number of regimes S is a parameter in Θ_r, and Θ_y is actually a set of different parameters for each regime, each indexed by Θ_y^s. This means that the SGP can be written as:

$$
\mathrm{sgp}(y_{1:(t-1)}, r_t, t;\ \Theta_y) = \sum_{s=1}^{S} f_{\mathrm{sgp}}(y_{1:(t-1)}, t;\ \Theta_y^s) \cdot r_t^s \tag{3.2}
$$

Note how each regime is weighted by the regime variable r_t^s. In the simplest case, this weight is binary ($r_t \in \{0,1\}^S$) – only one of the regimes is 'turned on', and all the others are 'turned off'. However, in some models, such as Smooth Transition, the weights can be continuous, with different regimes being partially 'on' at the same time.

Furthermore, Θ_y could encode different functional forms for each regime. As this is not common, I refer to f_{sgp} as the *SGP functional form*, or simply *SGP*, and the set of parameters Θ_y as the *regime nature*, as they define what actually changes between each regime. Figure 3.1 illustrates this structure.

Figure 3.1: The general RS DGP structure

$$
\text{DGP} \;=\; \underbrace{\overbrace{\text{SGP}}^{\sum_{s=1}^{S} \mathsf{sgp}(.\ ;\ \Theta_y^s)\cdot r_t^s}}_{\underbrace{\text{functional form}}_{f_{\mathsf{sgp}}} \;\&\; \underbrace{\text{regime nature}}_{\{\Theta_y^1, \dots, \Theta_y^s\}}} \;\&\; \overbrace{\text{RGP}}^{r_t = \mathsf{rgp}(.\ ;\ \Theta_r)}
$$

[2] Let $a : b := \{a, a+1, \dots, b\}$ for $a \le b \in \mathbb{Z}$, and $y_{a:b} := \{y_a, \dots, y_b\}$.

To construct a diverse set of DGPs, I combine different RGPs, SGPs, and regime natures. One of the challenges of this work is to choose a comprehensible set of these elements, and analyze their differences in a systematic but manageable way.

In the notation above I omitted the error term inside f_{sgp}. Many distributions are interesting, especially fat-tailed and skewed ones. The parameters Θ_y can even encode different error distributions across regimes. However, if we constrain the same distribution across regimes, with the possible exception of a multiplicative factor, we can simplify the notation and implementation, writing the DGP as a function that receives a sequence of random errors and returns the series and the regimes:

$$(y_{1:T},\ r_{1:T}) = \text{dgp}(\varepsilon_{1:T};\ \Theta_r, \Theta_y) \tag{3.3}$$

Consider the notation shorthand $y \coloneqq y_{1:T}$, and similarly for other variables, used for the rest of this work.

Let the set of considered DGPs be P (for 'processes'). These are present in the literature, as discussed in Section 2, and will be defined in Section 5.

3.2 Models

Consider a model mod as a function with (hyper-) parameters Θ_m that generates the fitted values and H-step ahead predictions of the series and regimes. The model can also return a set $\hat{\pi}$ of general metadata, e.g. the estimated coefficients.

$$(\hat{y},\ \hat{r},\ \hat{\pi}) = \text{mod}(y_{1:(T-H)}\ ;\ \Theta_m) \tag{3.4}$$

Notably, the number of estimated regimes $\hat{S}$ is a parameter in Θ_m, which may or may not be equal to S. Let the set of models be M (for 'models'). Also present in the literature, they will be defined in Section 5.1.

3.3 Regime conditional metrics

A regime-conditional (RC) metric met is function that receives a vector of series and a vector of regimes, and returns a sequence with one value for each regime. They are used to characterize the distribution of y_t or (y_t, y_{t-j}) within each regime.

$$\text{met} : (y, r) \mapsto \mathbb{R}^S \tag{3.5}$$

An example is the function that returns, for each regime s, the mean of the series weighted by r_t^s. This can be done for many common metrics, and is equivalent to mapping (y, r) to the S sets R_s of regimes' observations,[3] then applying the metric to each set. The benefit of the first approach is that it is more general, allowing for non-binary – i.e., smooth transition – regimes.

For the joint distribution (y_t, y_{t-j}), the metrics are more complex, as they must consider only the windows $(y_t, \dots, y_{t-j})$ fully contained in the same regime instance. This is further described in Apendix A.2.

In any case, RC metrics may lump together observations from different time windows. For them to describe a well-defined regime distribution, it is required that the series be stationary within each regime. This will impose restrictions on the DGPs that this work will consider.

[3] $R_s \coloneqq \{y_t\ :\ r_t^s = \max\{r_t\}\}$.

3.3.1 Within-regime stationarity

For the distribution of a regime to be well defined, all datapoints within it must be drawn from the same distribution. Formally, *within-regime strong stationarity* requires, for all regimes $s \in S$, a CDF F_s such that:

$$F_s(y_{t_1}, \dots y_{t_n}) = F_s(y_{t_1+j}, \dots y_{t_n+j}) \qquad \begin{array}{c} \forall \{y_{t_1}, \dots y_{t_n}\}, \{y_{t_1+j}, \dots y_{t_n+j}\} \subset R_s, \\ \forall j \in \mathbb{N}_* \end{array} \tag{3.6}$$

As we intend to characterize the distributions with specific metrics, weaker assumptions can be made. If we restrict ourselves to the moments of the (joint) distribution, we can require the weak version. Formally, *within-regime weak stationarity* requires,[4] for all $s \in S$, that:

$$\begin{array}{cc} E[y_t] = E[y_{t'}] & \forall y_t, y_{t'} \in R_s \\ Var[y_t] = Var[y_{t'}] & \forall y_t, y_{t'} \in R_s \\ Cov[y_t, y_{t-j}] = Cov[y_{t'}, y_{t'-j}] & \begin{array}{c} \forall \{y_t, \dots, y_{t-j}\}, \{y_{t'}, \dots, y_{t'-j}\} \subset R_s, \\ \forall j \in \mathbb{N}_* \end{array} \end{array} \tag{3.7}$$

Processes that have a non-binary r_t, i.e., smooth transitions, do not have truly separated regimes, and thus, generally do not satisfy the conditions above. Thus, the metrics cannot be interpreted as, e.g., "the mean of all datapoints in a regime". Still, their information might be useful, as will be studied in this work.

3.3.2 Aspects of RC metrics usage

There are two important aspects of the RC metrics usage. First is whether to use the whole sequence of values for each s, or to condense it into a single value of dispersion across regimes. An example of the latter is the 'average pairwise distance between the RC means', a single value that describes how distant the levels of the regimes are. This is equivalent to composing a dispersion function disp:

$$\mathrm{disp} \circ \mathrm{met} : (y, r) \mapsto \mathbb{R}^S \mapsto \mathbb{R} \tag{3.8}$$

Second, which series to use: the true or estimated ones. One can use the true values (y, r) to get the characteristics of the true DGP, and the estimated values $(\hat{y}, \hat{r})$ or $(y, \hat{r})$ to get the characteristics of the estimated model.[5] Another option is to calculate the difference between the former and the latter.[6] Another option is to calculate the metric of the difference $(y - \hat{y}, r)$ or $(y - \hat{y}, \hat{r})$.

Less generally, sometimes there are other possible estimators for the same population RC metric, instead of simply using $(\hat{y}, \hat{r})$. A special case is when the metric is a moment of the (joint) distribution, and the SGP is simple: one can simply plug the estimated parameters into the analytical formula for the moment, and generally have a better estimator. This is further discussed in Section A.2.

This framework allows for mixing and matching these options, each being useful to answer different questions. In this work, I use the estimated ones computed from $(y, \hat{r})$, as these are the values available to the econometrician in practice, while the true metrics are calculated analytically. Additionally, I

[4] The ACF condition should be read as "j'th autocorrelations between time-points within the same regime instance should always be equal", not as "j'th autocorrelations of every time-point in the same regime instance should always be equal", although the latter is true for AR processes with order higher than j.

[5] Note that the value of S and $\hat{S}$ can be different, and thus, so the dimension of the metric's output.

[6] This is only possible if $S = \hat{S}$ and there is an unambiguous way to match the estimated and true regimes.

ignore the metrics separated by regime, and condense their information by considering only their dispersion, as this is a simpler measure and more comparable across DGPs and models.

Let the set of metrics $(\text{disp} \circ \text{met})$ be C (for 'criteria'). These will be defined in Section 5.4, but are mostly based on the moments of y_t and of the pair (y_t, y_{t-j}), $j \in \mathbb{N}$, and the performance metrics for the dependent variable. I also discuss other interesting metrics that weren't considered.

One can also be interested in describing the RGP, with information such as the average duration of each regime instance, the transition probabilities and measures derived from it, amongst others. I'll use these as control variables in the regression analysis. Finally, regime-inconditional metrics can also be useful.

4 Simulation framework

One of the partial goals of this work was to create the theoretical framework described in the last section in a very general and expandable way, that easily allows for different exercises, even if they are not considered here. Similarly, the simulation structure was designed to follow the same concept.

There are the following steps to perform the simulations:

1. Generate random errors for all the DGPs.
2. For each DGP and simulation, fit and predict the model, generating (y, r).
3. For each DGP, simulation, and model, obtain $(\hat{y},\ \hat{r})$.
4. For each DGP, simulation, and model, compute each metric.
5. Aggregate the metrics, performance information, and DGP and model descriptors into a dataset.

4.1 Hyperparameters

For the forecast performance, I focus on 1-step ahead predictions. It would be interesting to expand that, be it with locally-projected models or not.

To obtain more than one prediction per simulation, I simulate a $T - H$-long series, and obtain H predictions. There are two possible approaches:

1. For each iteration $h \in 1 : H$, the model is estimated with the window $h : (T - H + h - 1)$, and generates $\hat{y}_{T-H+h}$.
2. The model is estimated once with the window $1 : (T - H)$, then for each h, $\hat{y}_{T-H+h}$ is generated using $y_{1:(T-H+h-1)}$.

The second approach is computationally cheaper, allowing for more simulations and DGPs to be considered. It is the one used in this work, but note that it is less accurate to what would be done in practice, as econometricians often re-estimate their models with new data.

Overall, the hyperparameters of the simulation are as follows:

- Number of simulations: I. Its main effect is on the the precision of the results, and diversity of series.
- Forecast horizon: H predictions of 1-step ahead values. Also affects the precision of the results, but does not change the diversity of series.
- Total number of observations: T. Its main effect is on the ability of the models to learn the dynamics and separate the regimes. Results for higher T's are more relevant for contexts with a lot of data, such as high-frequency financial data, while lower T's are more relevant for contexts with less data, such as macroeconomic data.
- Burn-in period: B. Its main effect is on reducing the dependence of the initial values, but with stationary processes, this is not too problematic.

Let $i \in 1 : I$, $I \in \mathbb{N}$ be the simulation index.

4.2 Simulation algorithm

I will only consider DGPs have the same error distribution – but note that a DGP can have a volatility parameter multiplying its error. For each DGP, indexed by $p \in 1 : |P|$, there are I random error vectors created, each of size T. Let E denote the set of all errors. Let $\mathrm{E}_{p,i}$ denote the vector of errors generated for the p-th DGP and the i-th simulation. Similar indexing definitions will be used for similar collections throughout this document.

Let Y and R denote the sets of generated series and regimes. They are computed given $\mathrm{E}_{p,s}$:

Initialize Y and R
for p **in** $1 : |P|$ **do**
  for $i = 1$ **to** I **do**
    $Y_{p,i},\ R_{p,i} \leftarrow P_p(\mathrm{E}_{p,i})$
  end for
end for

Now, for each simulation, I estimate each model, generating the sets $\hat{Y}$, $\hat{R}$, and $\hat{\Pi}$. The models are trained using only $y_{(B+1):(T-H)}$, to avoid the burn-in period and leave space for the forecast horizon. The nesting order is the same, but with an additional inner loop for the model estimation.

Initialize $\hat{Y}$ and $\hat{R}$
for p **in** $1 : |P|$ **do**
  for i **in** $1 : I$ **do**
    for m **in** $1 : |M|$ **do**
      $\hat{Y}_{p,i,m},\ \hat{R}_{p,i,m},\ \hat{\Pi}_{p,i,m} \leftarrow M_m(Y_{p,i},\ R_{p,i})$
    end for
  end for
end for

Then, for each model, the dispersion of the RC metrics are calculated and stored as columns of a dataset D. Each row of D is identified by (p, i, m).

Initialize D
for (p, i, m) **in** $(1 : |P|) \times (1 : I) \times (1 : |M|)$ **do**
  for c **in** $1 : |C|$ **do**
    $D_{(p,i,m),\ c} \leftarrow C_c(Y_{p,i,m}, R_{p,i,m},\ \hat{Y}_{p,i,m},\ \hat{R}_{p,i,m})$
  end for
  $\hat{\Pi}_{p,i,m}$ is appended to $D_{p,i,m}$.
  Categorical variables (p, i, m) are appended to $D_{p,i,m}$.
end for

Recall the discussion in Section 3.3 about the two different aspects of RC metrics usage. With different options, the function C_c can use different inputs (Y, R, $Y, \hat{R}$, or $\hat{Y}, \hat{R}$), which is represented by all four objects being passed to it. Additionally, the function could return the whole sequence of RC metrics, not a single value, then, each row would be identified by (p, i, m, s). While the metrics recieve the full dataset, estimation is only done with $(B + 1) : (T - H)$.

The dataset D is already in a friendly format for analyzing the relationship between the performance of each observation and the characteristics of the regimes, as well of considering stratifications by DGP and model.

5 Implementation

The framework described in the last two sections is very general, and allows for a lot of different exercises. In this specific work, I focus on a specific set of DGPs, models, and metrics. These are described here.

First of all, I focus only on DGPs where there are regime switching, and specifically two regimes ($S = 2$). More information about the ability of the models to identify regime dynamics with different (or zero) number of regimes is an interesting topic. A no-RS model is included as a baseline.

The choices of hyperparametrization were made to balance the 'population' of DGPs. The SGP was restricted to a simple stationary $AR(1)$ process. I consider the Markov Switching, Self-Exciting Threshold, and Smooth Transition RGPs, each with equal representation and with symmetric and asymmetric variations. There are two RNs for each of the three $AR(1)$ parameters, a big and a small change. The related hyperparametrizations were chosen guided by the concept of 'regime separation', described in Section 6. Two additional models are included: K-Means and Random Forest.

The choice of models and their hyperparametrization is more flexible, as they do not affect the 'population' of the experiments. Each of the RGPs' empirical model counterparts is used, with a 'generic' hyperparametrization. But it would be interesting to increase the diversity of models.

The metrics are limited to the most essential descriptors of the regime distributions, the 1st, 2nd moments, and the lag 1 autocorrelation. This is another set that could be expanded easily. Performance and RGP-related metrics were also defined for the regression analysis.

Finally, some diagnostics on the series generation and model estimation are discussed, as well as describing the final dataset.

5.1 Considered SGPs

The functional form of the SGP could be important in its interaction with the other ingredients of the DGP. Additionally, some topics are interested in specific SGPs, such as conditional volatility in finance and GARCH models. For now, however, this does not seem to be the main point of interest. I will consider only an $AR(1)$ process, for its simplicity, popularity, and ease of estimation.

Additionally, I'll only consider a Gaussian distribution for the error term, ignoring fat-tailed and skewed distributions. The distribution is regime-invariant, except for the multiplicative variance parameter σ.

As discussed, it is useful to consider only within-regime weak stationarity, even though many interesting DGPs are non-stationary. This restricts the absolute value of the $AR(1)$ parameter to be less than 1. The only SGP functional form considered is the following:

$$\begin{aligned} & f_{\text{sgp}}(\cdot\ ;\ (\mu^s, \rho_1^s, \sigma^s)) = \mu^s + \rho_1^s y_{t-1} + \sigma^s \cdot \varepsilon_t \\ & \varepsilon_t \sim \mathcal{N}(0,1) \\ & |\rho_1^s| < 1, \quad \sigma^s > 0, \quad \forall s \in 1:S \end{aligned} \tag{5.1}$$

Several others SGP's could be considered, such as ones with transformations of y_t as regressors, non-linear regression forms, or even decision trees, as in the common model Markov-switching Random Forest. Still, the $AR(1)$ is an essential building block, and its simplicity helps isolate the effects of the other ingredients.

5.2 Considered RGPs and models

The next 'ingredient' is the RGP. I will consider the options Self-Exciting Threshold (SET), Smooth-Transition (ST), and Markov-Switching (MS). No regime switching (noRS) is included as a benchmark.[7]

Each of these RGPs has empirical model counterparts, which are also considered.

The formal definition of each RGP/model is presented in Section A, first the RGP hypothesis, then the empirical model's estimation strategy.

For all RGPs, an option with equally likely regimes and an asymmetric variation is considered.

- **No Regime Switching:**
  - Always in regime 1.
- **Self Exciting Threshold:**
  - Fixed hyperparameters: switching based on y_{t-1}. Different lags are often specific to timing-related issues, and not considered here.
  - A single threshold at 0.5, and a single threshold at 0.9.
- **Smooth Transition:**
  - Fixed hyperparameters: switching based on y_{t-1}, logistic's CDF as transition function.
  - A single threshold at 0.5, and a single threshold at 0.9.
- **Markov Switching:**
  - Symmetric matrix, high persistence ($P(s|s) = 0.9$).
  - Asymmetric matrix, high persistence ($P(1|1) = 0.9$, $P(1|2) = 0.3$).

A Structural breaks model was considered, and all analisies pointed to an overall imcomparable behavior with the other models, as it does not have reocurring regimes, as expected. Thus, it was dropped from the final set of models.

The values were chosen to generate a 50% proportion of regime 1 in the symmetric case, and 75% in the asymmetric case for each DGP. The Table 5.1 shows the absolute difference between the regime 1 proportion and 0.5, that is, we would like to have values close to 0 for the symmetric case, and close to 0.25 for the asymmetric case. Note that the RGP and RN interact, in such a way that the same RGP can generate different regime proportions for different RNs.

Table 5.1: Proportion of regimes across DGPs

	Uncond.	μ $(0, 0.5)$	μ $(0, 1)$	ρ_1 $(0.1, 0.9)$	ρ_1 $(0.4, 0.6)$	σ $(1, 1.5)$	σ $(1, 2)$
No RS	.50	.50	.50	.50	.50	.50	.50
MS, $p_{21} = 0.1$	.02	.03	.02	.03	.02	.03	.02
MS, $p_{21} = 0.3$	.23	.24	.23	.23	.23	.24	.23
SET, $\tau = 0.5$	.12	00	.24	.12	.01	.18	.19
SET, $\tau = 0.9$	.19	.18	.04	.25	.15	.28	.27
ST, $\tau = 0.5$	.13	00	.13	.19	.25	.11	.08
ST, $\tau = 0.9$	.19	.12	.02	.30	.34	.21	.16

[7] A structural breaks model was considered as a benchmanrk of model without reocurring regimes, but was ultimately incomparable with the other models.

Two aditional models, with no RGP counterpart, are considered: random forest (RF) and K-Means clustering KM. Both recieve four lags of the series, and the rolling-average, ACF(1) and SD of the series. The RF does not generate a regime series, and is included as benchmark in only some of the analyses. The RF uses 50 trees and 10 min. terminal node size.

For the models, most hyperparameters are as follows:

- All the coefficients are assumed to change across regimes, as this is a common assumption, especially in the face of possible mis-specification.
- The number of regimes $\hat{S}$ is fixed, not estimated. Models are estimated with 2 regimes.
- The values of model-specific hyperparameters are the same as the related RGP's values.

5.3 Considered regime natures

The following regime natures are considered, each representing a different way in which the SGP parameters change across regimes:

- **Mean (μ) change:**
  - Small difference: ($\mu^1 = 0$, $\mu^2 = 0.5$)
  - Large difference: ($\mu^1 = 0$, $\mu^2 = 1$)
- **Persistence (ρ_1) change:**
  - Small difference: ($\rho_1^1 = 0.4$, $\rho_1^2 = 0.6$)
  - Large difference: ($\rho_1^1 = 0.2$, $\rho_1^2 = 0.8$)
- **Volatility (σ) change:**
  - Small difference: ($\sigma^1 = 1$, $\sigma^2 = 1.5$)
  - Large difference: ($\sigma^1 = 1$, $\sigma^2 = 2$)

Note that the regimes are always ordered increasingly by the parameter of interest. In the asymmetric RGPs, the rarer regime is always the second one, with the higher value of the relevant parameter.

The values were chosen in accordance with the regimes proportion discussed in the last section, and also to generate a reasonable level of regime separation, as described in Section 6.

5.4 Considered metrics

The goal with RC metrics is to capture the change in the series characteristics across regimes. One important option is the estimated parameters of the model for each regime, e.g., $(\hat{\rho}_s)_{s\in\hat{S}}$, $(\hat{\mu}_s)_{s\in\hat{S}}$, etc. One might think that this would outshine all other metrics, but in more complex cases where more than one parameter changes, this becomes less useful. More general metrics generate benefits from their abstraction over the DGP. Additionally, in simple SGPs, there often is a metric that is directly connected to changes in parameters, such as the conditional average for changes in intercept.

In this work, I focus on the moments of the distribution of y_t and (y_t, y_{t-j}). Specifically, the RC metrics considered are the RC mean, RC standard deviation, and RC autocorrelation of lag 1. Higher lags could be considered, but in the simple $AR(1)$ context this would bring little additional information.

As stated before, the RC mean and RC SD are simply the mean and SD of each set R_s. The autocorrelation is similar, but must be calculated separately for each concurrent set of observations in R_s. The formal definitions are stated in the Section A.2.

As the focus is on the dispersion of RC metrics, two important measures to consider are the standard deviation and the average pairwise absolute difference. For only two regimes, they are very similar and the absolute difference is more intuitive. All the metrics are composed such that all $\text{disp} \circ \text{met} \in C$ return a single real value, and $\text{disp}(x) = |x_1 - x_2|$.

There are some possible expansions on this work's metrics calculation. One is to use non-standard weights for the empirical moments, giving more importance to observations near the edges of regimes' instances. Another is to use a cluster separation measure, such as the silhouette score, instead of a simple absolute distance between the RC metrics. Finally, one can use distribution distance metrics, such as the Earth Mover's Distance, on the empirical distribution of each regime. These are not currently considered.

No regime-unconditional metrics are considered. The list of considered metrics is as below:

- First moment: RC mean $\hat{\mu}(y|S)$.
- Second moment: RC standard deviation $\hat{\sigma}(y|S)$.
- First autocorrelation: RC 1st autocorrelation $\hat{\rho}_1(y|S)$.

5.4.1 Performance and RGP metrics

The performance metrics considered is the RMSE for forecasting performance. The MSE is not included, following Dacco; Satchell (1999). The fit performance is measured by R^2 for y and binary mean error for r.

Other metrics pertaining to the RGP will be included in the regression analysis: the number of regime switches divided by T, as a measure of switching frequency; the absolute difference between the average duration of regime 1's instances and regime 2's instances, as a measure of regime asymmetry.

For works with more than two regimes, more complex measures of regime asymmetry can be used, ones that consider the whole matrix of transition probabilities.

5.5 Simulation and diagnostics

The implementation of the simulations, as well as their analysis in the next sections, is done with the R programming language, and the code can be found in this paper's repository. The code is highly modular and fully reproducible, following the intent of setting up an expandable framework.

Following Section 4.1, the chosen hyperparameters are as below. Some values are lower than they could be due to computational constraints. The choice of T is discussed in Section 6.2.

- Number of simulations: $I = 500$.
- Forecast horizon: $H = 10$ predictions of 1-step ahead values.
- Total number of observations: $T = 100$.
- Burn-in period: $B = 4$.
- As described above, there are 6 RNs, 7 RGPs, and 4 models.

The error sequences were generated in parallel, using `rTRNG::rnorm_trng`. The models were estimated with `stats::lm`, `mbreaks::dofix`, `tsDyn::setar`, `tsDyn::lstar`, `MSwM::msmFit`, and `randomforest::randomForest`.

On top of visualizing the series and guaranteeing no missing values, some diagnostics on the simulation, model estimation, and metrics calculation are performed. All of the diagnostics are presented in Section B.

The errors should be i.i.d. Gaussian with mean 0 and should not present any pattern, especially across the parallelization structure. This is guaranteed by the TRNG library, but it is also checked.

Some observations had to be removed. Table 5.2 lists the reasons and the amounts, some observations have multiple issues thus the difference between the '% Bad' and '% Removed' columns. Some models did not converge and produced no output. Others couldn't estimate some parameters. The need to remove those is straightforward.

Some models' predictions were dominated by one regime and produced zero or only one observation of the other. While this is not a failed estimation, calculating the dispersion of regimes distributions is impossible in these cases, and they would be removed from most analysis anyway. Section B.3 plots the distribution of regimes' proportion, and notably, K-Means has the most balanced regimes.

The last removal is less straightforward. Some models generated unreasonably large errors, and parameters unreasonably outside the normal range, so they would commonly be disregarded. This is more subjective and prone to cherry-picking; thus I was parsimonious in this removal. Only RMSEs higher than 50, means higher than the 90th quantile of the whole dataset, and ρ_1s higher than 10 standard deviations of all the ρ_1 estimates were removed.

Section B.2 shows the distribution of fits and RMSEs, notably, the RF has signs of overfitting, which is because the chosen 'generic' hyperparametrization is overkill for the simple SGPs considered. Section B.4 shows the distribution of the estimated AR parameters, notably, ST has distributions with more variance. Section B.4 shows distribution of other paramters of the models on models. The RF importance ranks the first two lags then the rolling average as the most important, while the ACF and SD have much lower importance.

Table 5.2: Estimation issues

	Bad obs.	% Bad	% Removed	Obs. left
Full sample	0	0%	0%	126,000
Bad convergence	148	0.1%	0.1%	125,852
NAs in parameters	245	0.2%	0.2%	125,607
Low #obs. in regimes	967	0.8%	0.6%	124,859
Unreasonable parameters	1,803	1.4%	1.3%	123,162
Unreasonable errors	145	0.1%	0%	123,137
Total	**3,308**	**2.6%**	**2.3%**	**123,137**

The average coefficient generated by the models is presented in Section B.4. Only the matched model-RGP pairs are included, and a test of difference agains the true parameter is performed. Many tests don't pass, but that is expected, as all parameters are allowed to change across regimes. It is important to note that they generally do (except for σ).

To consider the metrics estimation, the regime-conditional and unconditional moments are estimated using and tested against their true values in Section B.5. The estimations use (y, r), while the true values are calculated via the analytical formula (the unconditional true values are not calculated). Again, the tests are not generally expected to pass, specially given their high power.

As a final placebo test, the forecast performance (RMSE) was regressed against the index i of the simulation. The Table 5.3 shows generally no relation, as expected.

Table 5.3: RMSE and simulation index relationship

	RMSE
i	-0.346 (4.293)
i^2	1.578 (2.036)
i^3	0.553 (1.444)
$\log(i)$	0.004 (0.014)
Constant	1.284^{***} (0.073)
Observations	123,137
R^2	0.00002
Adjusted R^2	-0.00001
Residual Std. Error	1.026
F Statistic p-value	0.622
Note:	*p<0.1; **p<0.05; ***p<0.01

6 Regimes' distribution

The goal of this section is to explore how the information of the regime distributions, as captured by the RC metrics, tells us about the DGPs and models. This information will be useful to put the systematic results of the next section into perspective, and also present some stylized facts for the econometrician.

The first two sections, 6.1 and 6.2 summarize, for each DGP, how separated the regimes are in terms of each metric, considering the whole sample, and then varying the sample size. Section 6.3 explores how the models capture the regime separation.

6.1 Regime separation in T

The first step is to understand what each DGP implies for the distribution of y_t across regimes. A direct way to do this would be to plot the regime-conditional distributions (and, ideally, the joint distribution of (y_t, y_{t-1})). However, those visualizations quickly become hard to digest when repeated across many DGPs and hyperparameters.

The approach in this work is instead to characterize each regime distribution with regime-conditional metrics, and to summarize "how different the regimes are" by a dispersion across regimes. As stated before, more metrics than the ones considered here could be necessary to fully capture the differences between regimes.

The first object of this subsection is the Table 6.1. It should be read as a compact 'profile' of the DGP in terms of regime separation. Each *row* corresponds to one DGP configuration, grouped by the regime generating process (RGP) and the regime nature's parameter (RN); For each RC metric (mean, lag-1 autocorrelation, and standard deviation), there are two *columns* corresponding to the 'small' and 'big' parameter changes of the RN; Each *cell* is the absolute difference of the corresponding RC metric across the two regimes, as well as the '(SD)' and non-zero test p-value's stars. The table uses only symmetric RGPs, as well as 100 of the simulation indexes. The metrics are calculated with (y, r).

Table 6.1: Regimes' metrics separation across DGPs

		$d(\hat{\mu}(.))$		$d(\hat{\rho}_1(.))$		$d(\hat{\sigma}(.))$	
RGP	RN	$(0, 0.5)$	$(0, 1)$	$(0.1, 0.9)$	$(0.4, 0.6)$	$(1, 1.5)$	$(1, 2)$
MS	μ	.85*** (.02)	1.62*** (.03)	.01 (.01)	.01 (.01)	.02 (0)	.02 (.01)
	ρ_1	.03 (.06)	.07 (.3)	.11*** (0)	.38*** (.05)	.12*** (.01)	.53*** (.15)
	σ	.03 (.1)	.01 (.15)	.05** (0)	0 (.03)	.48*** (.06)	.99*** (.14)
SET	μ	1.52*** (.04)	2.29*** (.1)	0 (.01)	.22*** (.05)	.01 (0)	.05*** (.03)
	ρ_1	1.02*** (.05)	1.3*** (.19)	.02 (.03)	.22*** (.02)	.02 (.06)	.22*** (.06)
	σ	1.09*** (.17)	1.24*** (.3)	.13*** (.05)	.14*** (.06)	.49*** (.1)	.98*** (.16)
ST	μ	.64*** (0)	.47*** (.03)	0 (.01)	.08*** (.03)	0 (.01)	.03** (.01)
	ρ_1	.92*** (0)	1.06*** (.07)	.21*** (.02)	.41*** (.05)	.07*** (.01)	.22*** (.02)
	σ	1.15*** (.01)	1.41*** (.07)	.08*** (0)	.05*** (0)	.28*** (.01)	.58*** (.07)

Note: *p<0.1; **p<0.05; ***p<0.01

In the first line, MS and μ change, we can see that the average separates the regimes, and only it, as would be expected. Then, with a change in ρ_1, we can see that both the ACF and the SD differentiate the regimes. This is because the standard deviation depends on ρ_1. When σ changes, only SD differentiates the regimes. This creates a mapping that connects which estimated metric shows differentiation into which parameter is changing in the DGP. This is not a silver bullet, as this mapping will not be the same as we consider other RGPs.

MS has the cleanest result because its RGP doesn't depend directly on y, so it interacts less with the RN. For (SET, μ), the average is separated, but the (big) ACF and SD also separate. This happens because in the 'big' regime a higher average creates a feedback loop that keeps values higher. With a ρ_1 change, the average also increases: in the 'big' regime higher ρ_1 amplifies large values, raising the average. For σ, a similar effect occurs: the 'big' regime has larger variation and larger maxima, while the 'small' regime has smaller variation and smaller minima, increasing the average difference across regimes.

With ST, we have a similar result, as its RGP is very similar to the SET. Notably, ST shows higher separation in the 'small' RNs, which SET does not show significantly. Overall, this table exemplifies the importance of the RGP–RN interaction. This is relevant for understanding models' performance and how metrics' profiles vary with it.

6.2 Regime separation across t

It is intuitive that the sample size is an important factor in the ability of the estimated metrics to denote separations in the regimes. The ability to learn regime dynamics depends not only on which parameter changes (RN), but also on how quickly the induced regime separation becomes statistically visible as T increases. To study the interaction between RGP, RN and T, I calculate each metric with the data up to each time-point $(1:2,\ 1:3,\ ...,\ 1:T)$. By graphing the last time-point considered on the x-axis, and the difference of the RC metric between regimes on the y-axis, we can see how the separation evolves across sample size.

The figures 6.1, 6.2, and 6.3 present the results. The x-axis is the last time point included in the 'effective sample', used to calculate the line and ribbon at that x value; each line is the average (across simulations) of the metrics' dispersion, and the ribbon is calculated with the standard error (across simulations) times 1.96. Both are stratified by RGP symmetry, indicated via color. The graphs use only the big RNs, and 20 simulation indexes. The metrics are calculated with (y, r).

We have two main results. The first is the effect of asymmetry. The average values across asymmetric and symmetric cases are very similar, but the standard deviation is not. This is because the standard deviation is constrained by the small amount of observations in the less frequent regime, and this number grows very slowly across time.

The second result is that the values and their standard deviations seem to be more or less stabilized around $T = 60$, which means that for a series of this size, between 60 and 100, which is the observation size of this work, we possibly would have similar results on the usefulness of these metrics. But for series smaller than that, the metrics would be significantly less informative.

6.3 Models and regimes

To study how the models estimation relate to the regime separation, one could compare the estimated metrics with the true ones. This is done in Figure 6.4, where the absolute difference between the true and estimated RC metric's dispersion. Each panel (row) is a parameter change, and (column) a metric. The estimated metrics are calculated with $(y, \hat{r})$, while the true metrics via the analytical formula, thus, there is the composition of the metric estimator error and the $\hat{r}$ error. The metrics were scaled by their MAD for comparability.

Figure 6.1: Regime separation - MS

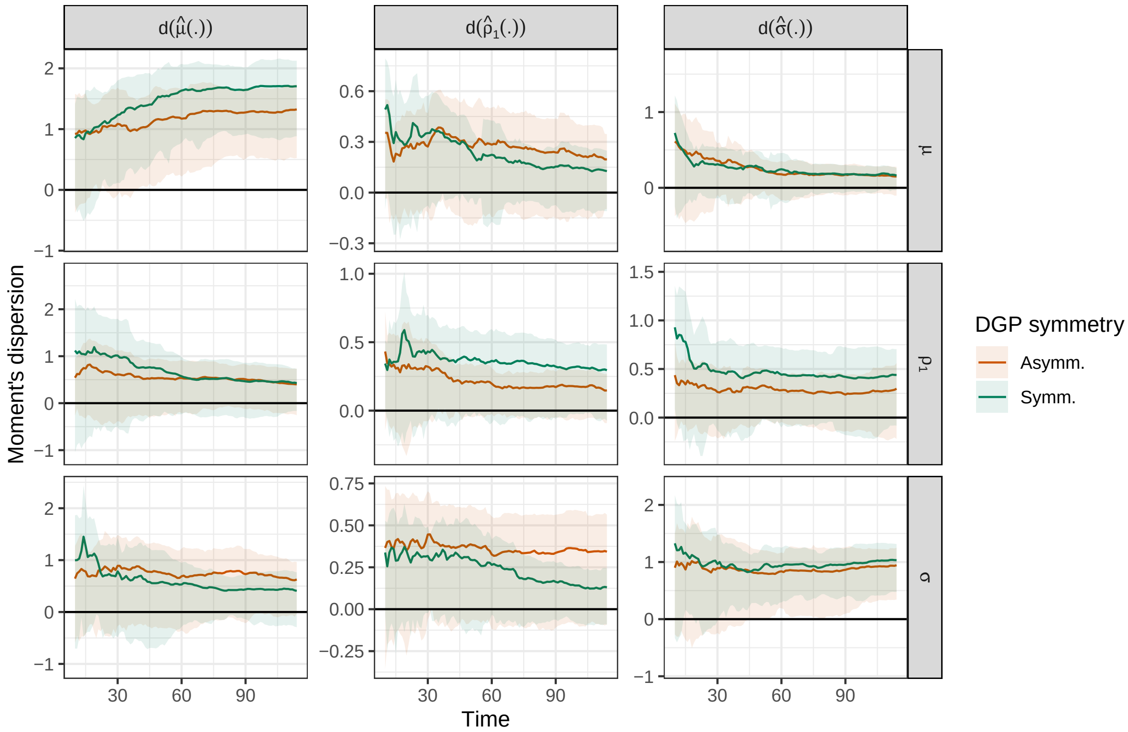


Figure 6.2: Regime separation - SET

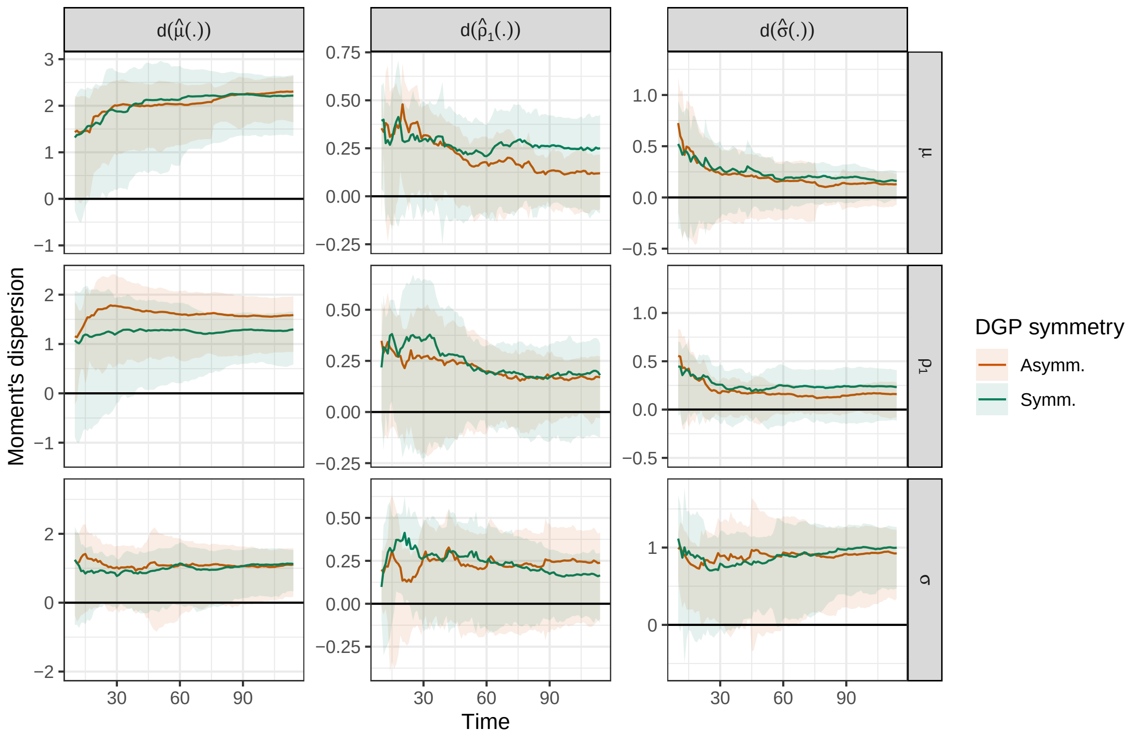


Figure 6.3: Regime separation - ST

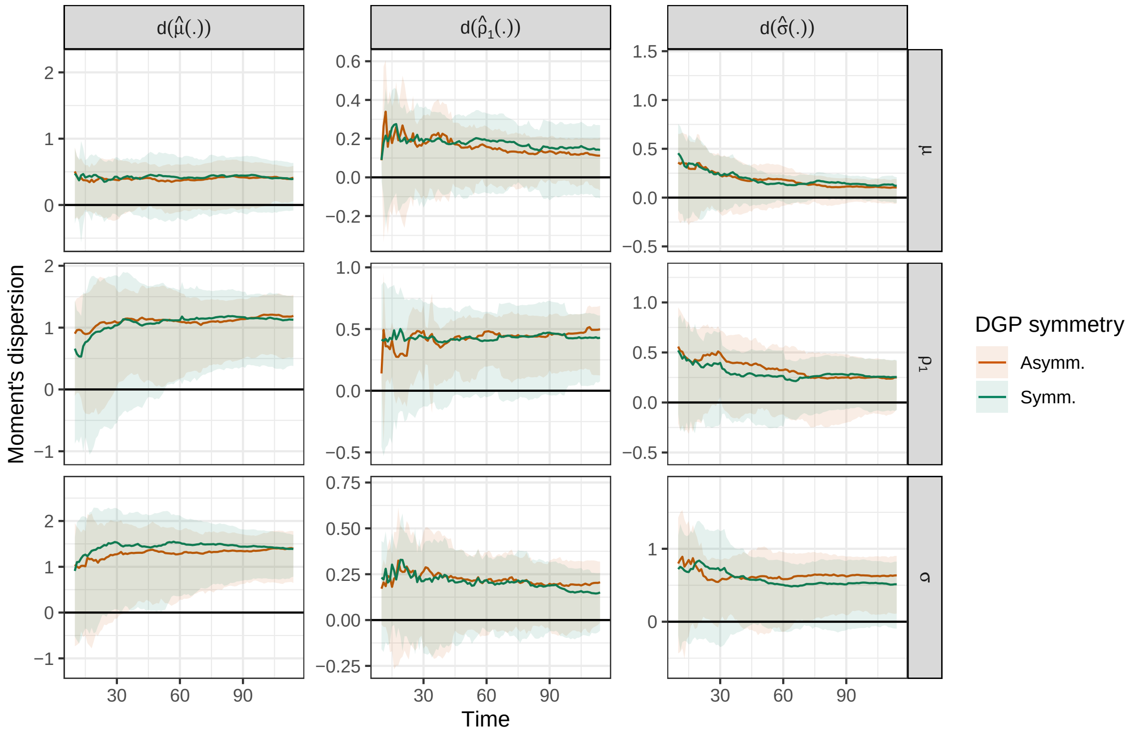
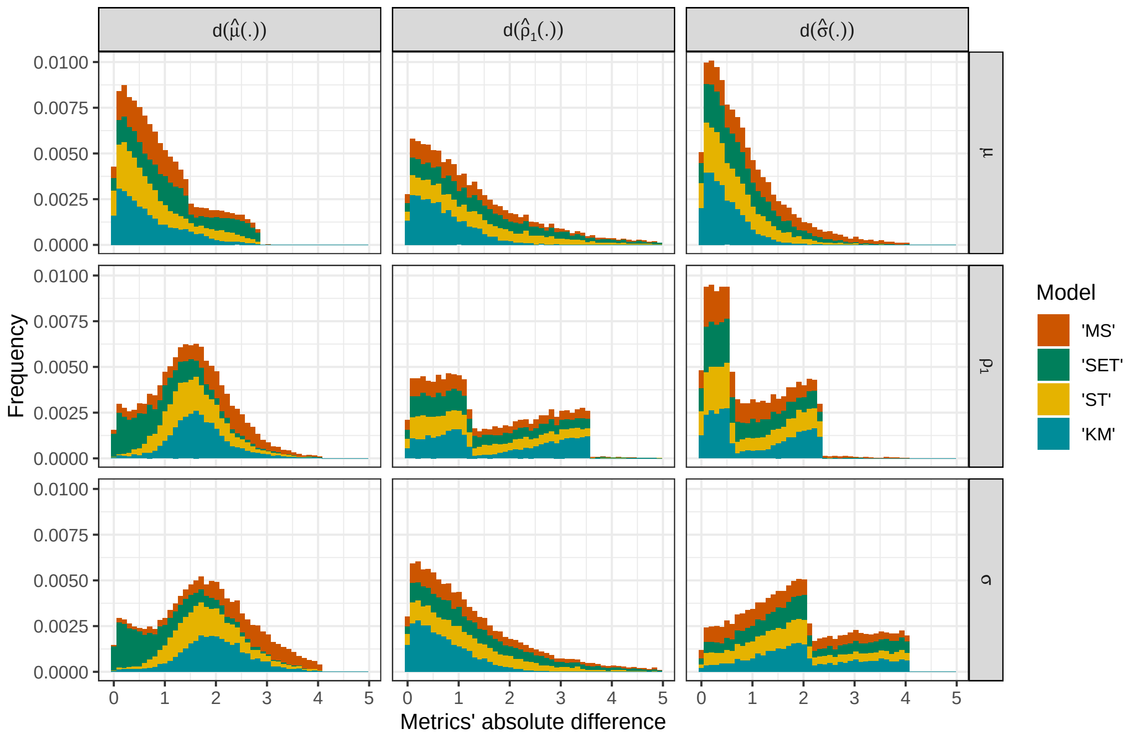


Figure 6.4: Metrics difference



We can see that KM and ST do well with average when μ is changing, but not otherwise, while SET has a more balanced result, and MS being the clear last. KM overall shines with μ changes, possibly for having 4 lags of the (level of) the series in its composition. More importantly, the distribution for any given metric varies widely across RNs, this means that no model makes a 'fixed' error for each metric, depending on the unobservable RN, the correctness of the estimated metrics will vary. In opposition, a graph stratified by RGP would show that each metric error is somewhat invariant across RGPs. This will be relevant in Section 7.2. The breaks in the distributions comes from the fact that the true metrics are set values for each DGP.

Another usefull analysis comes from relating the model output (estimated fit, r, parameters) to the performance of the models. The fit and parameters, continuous variables, are studied in Section 7.3. The binarized regime assignment has special importance, as it is directly related to regime separation, and can be used to split the forecasting errors into "correctly identified" and "incorrectly identified" observations. The figures 6.5-6.8 compare the distribution of individual forecasting errors, conditional on whether the underlying regime was correctly identified. Each panel corresponds to a DGP configuration, and presents the distribution of forecasting stratified by regime correctness; Each figure corresponds to one model. The graphs are done with big RNs, symmetric RGPs, and 100 simulation indexes.

In general terms, the 'correct' distributions have slimmer tails, as would be in line with the literature. But it is important to note it is not always the case, especially for the MS model, a result that will be discussed in Section 7.1. For the ST model with no-RS RGP, we have overall high errors with bi-modal distributions, which will be discussed in Section 7.2.

Figure 6.5: Regime and series prediction - no-RS

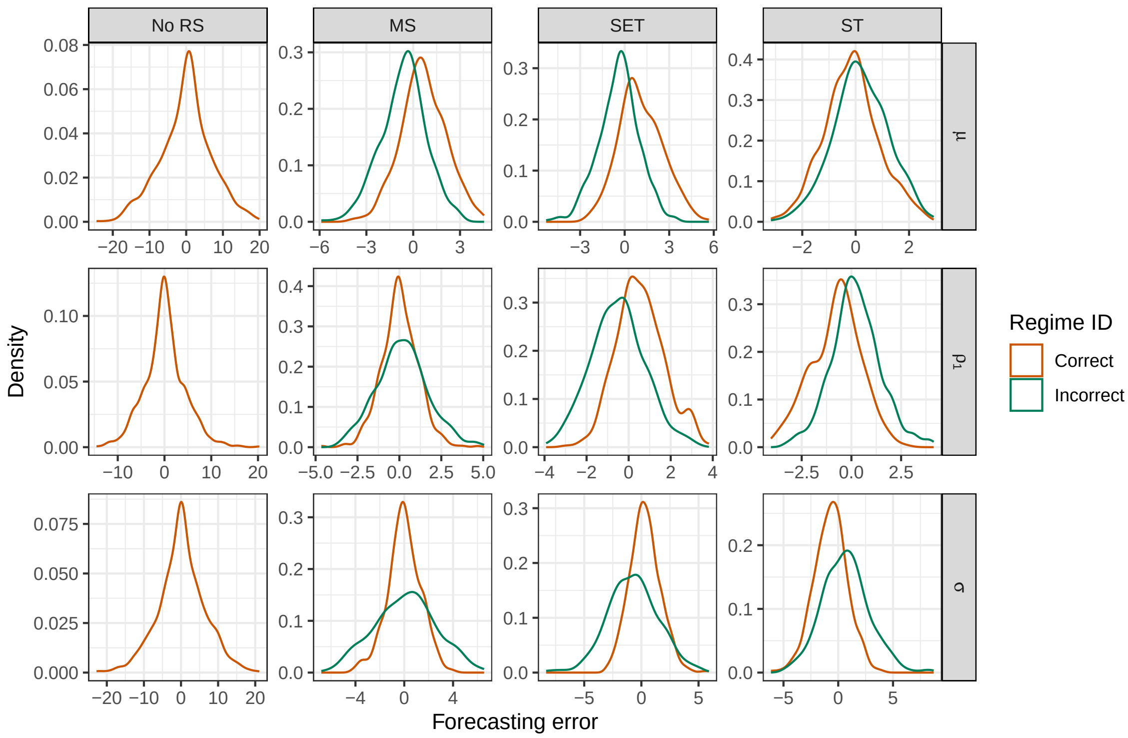


Figure 6.6: Regime and series prediction - MS

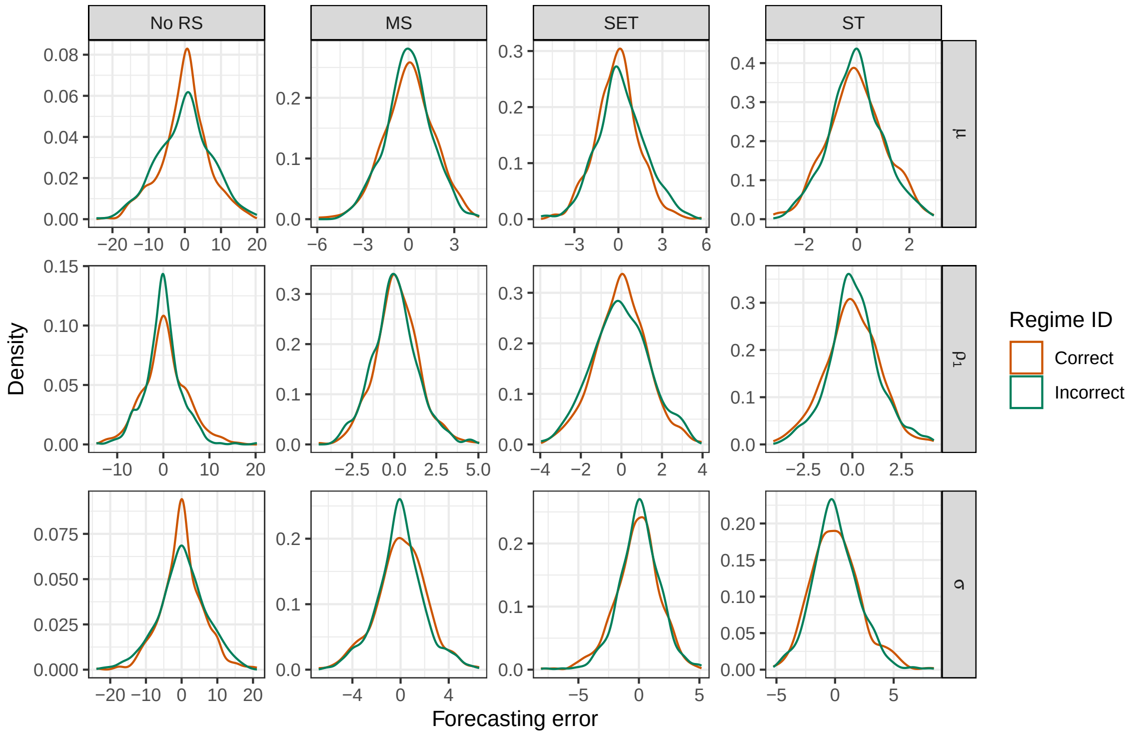


Figure 6.7: Regime and series prediction - SET

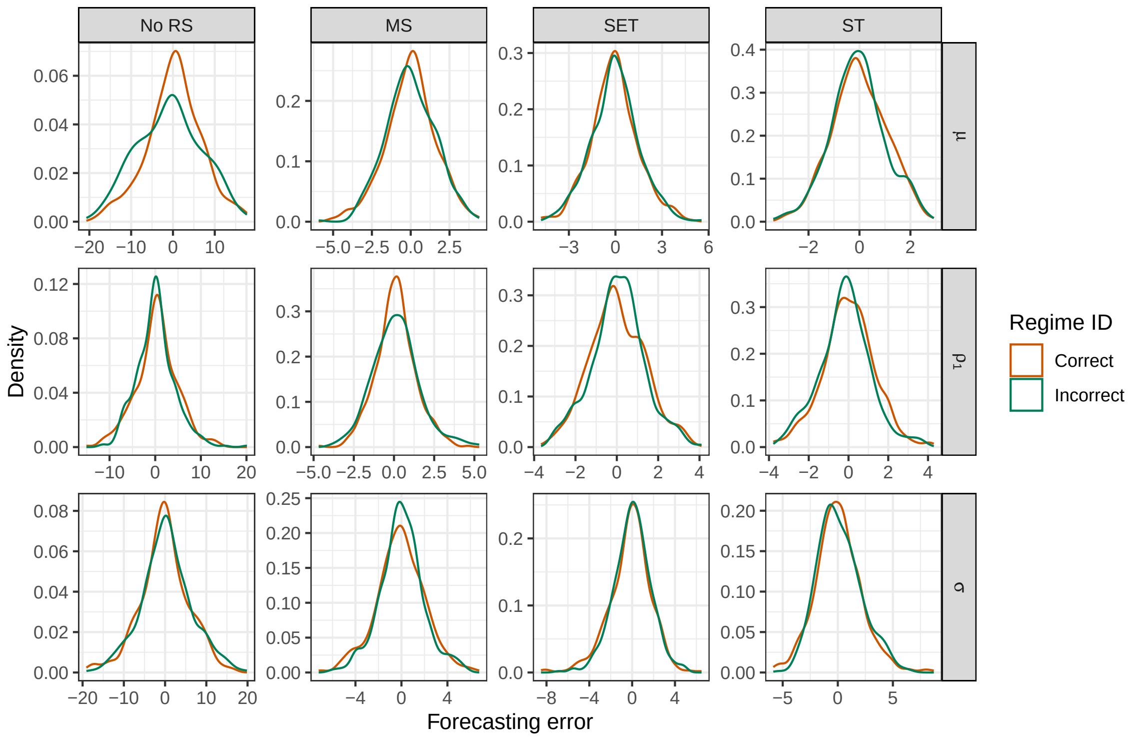


Figure 6.8: Regime and series prediction - ST

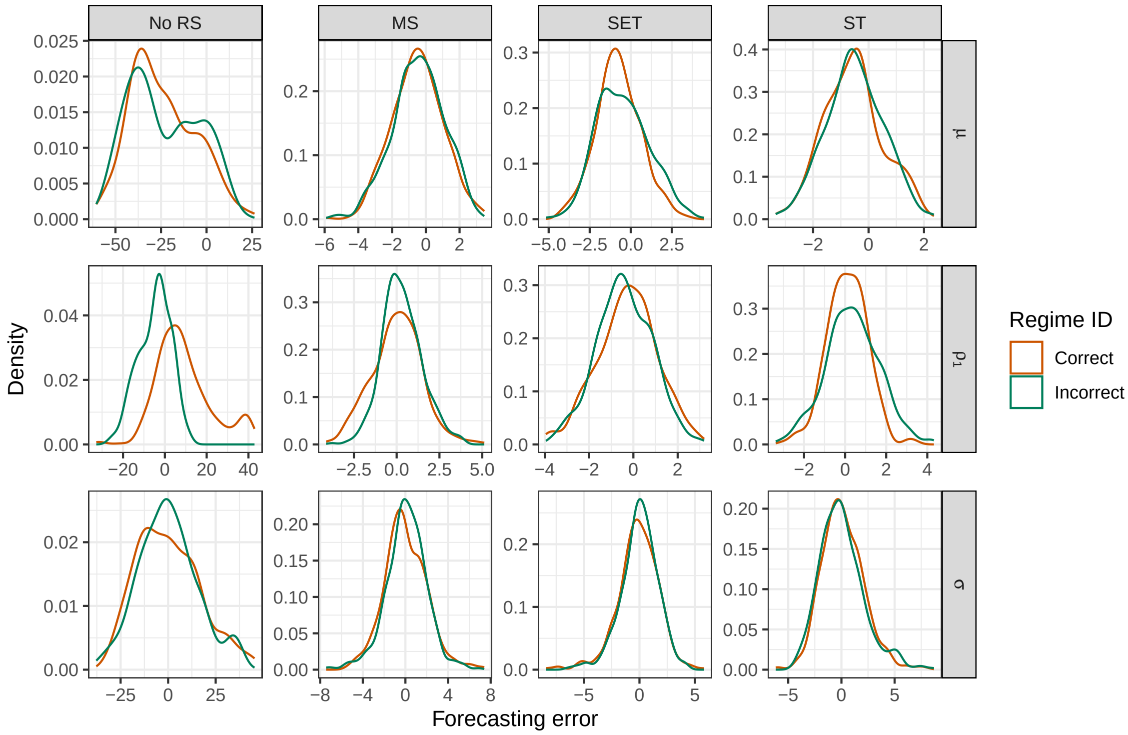
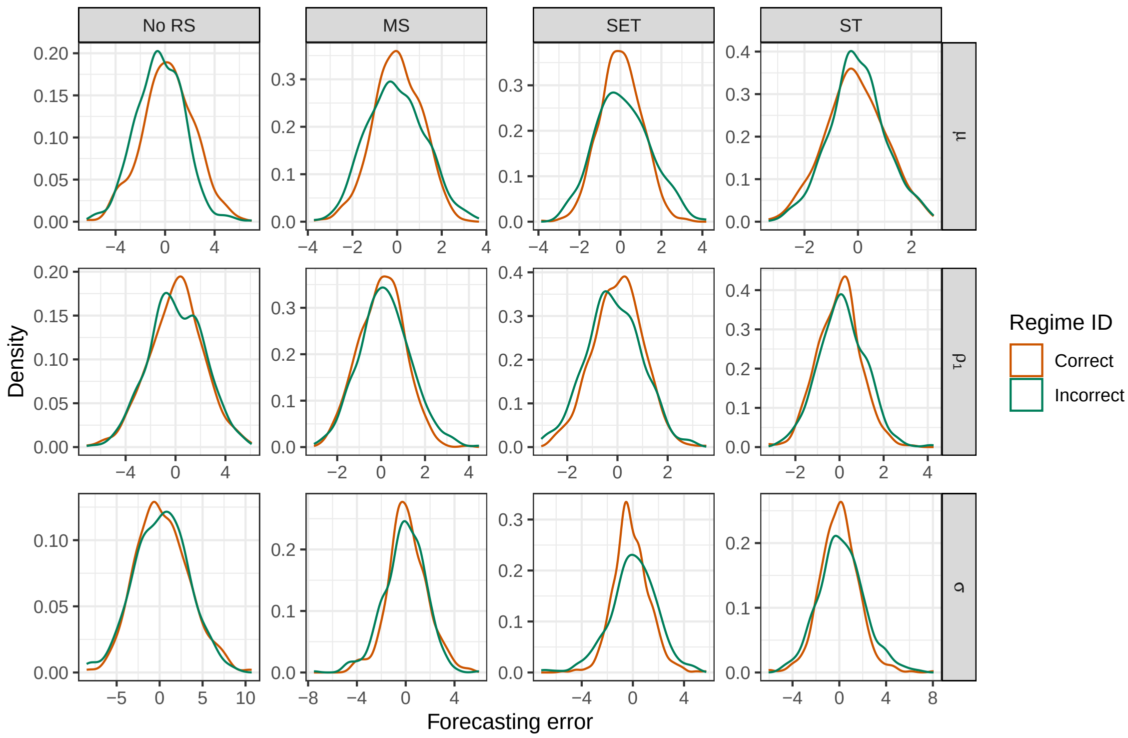


Figure 6.9: Regime and series prediction - KM



7 Performance analysis

In this section, the goal is to obtain systematic results on the performance of models, and how the regimes' distributions might affect and inform performance. First, I analyze the overall performance of each model through their fixed effects. I (i) add controls to understand which component of the models is most related to their performance; and (ii) stratify the DGPs to understand how performance changes across scenarios.

I continue studying the models' performance by further analyzing their interaction with the DGPs. I (i) analyze the effect of model-RGP mismach, and interact it with the DGP options, to see if the mismatch effect change across scenarios; (ii) analyze specific model-DGP pairs; and (iii) consider a different way to describe the DGPs, in terms of the RC metrics that they generate, and check how each model fares against each profile of series. I consider if any results induce a practical recommendation for the econometrician.

Then, in the topic of mis-specification, I test the hypothesis that underestimating the number of regimes is less harmful when the regmes' distributions are less separated, and if the opposite is true for overestimation.

Finally, I analyzed which model component (estimated fit, r, parameters, or metrics) is more associated with each models' performance. For each model, there are different points of interest for the econometrician.

All the regressions have RMSE as the dependent variable, so higher coefficient values imply a worse performance associated with the given variable. The metrics and parameters are normalized $|x - \text{median}(x)|/\text{mad}(x)$, except for the RMSE. Some metrics are not available for all observations, such as the ACF that requires at least one length-2 instance of each regime in the series, thus the number of observations in each regression can vary. A Δ symbol means the absolute difference between the estimated value (often with $y, \hat{r}$) and the analytical true value.

The results, some more than others, are conditional in the population of DGPs considered, and should not be interpreted universal truths for any serie.

7.1 Models' fixed effects

The fixed effects are presented in Table 7.1. This excludes the no-RS and RF model. The first column, without controls, indicates that SET and ST perform similarly and better than MS. This can be due to the higher prevalence of threshold-based DGPs than Markov-based ones in the DGP pool. All of them have significantly worse results than KM.

To understand how the qualities of the models compose each of these fixed effects, let's add controls for matching the true vs. estimated fit, r,[8] parameters, and metrics' dispersion, then remove them one by one. A better FE without a control means that relationship between the model and that control is a positive component in its FE. Comparing (2) and (3), MS has the highest improvement, given its flexible structure that can fit very general data, compared to the rigidity of the threshold-based models.

With Columns (2) and (4), we can see the counterpart effect: MS has a hard time matching the regimes, given its geometric distribution that is often not in line with threshold-based regimes. ST is slightly worse than SET, but this is because the regimes measure is binarized and does not utilize the continuity of ST regimes. The only one with a positive relationship is KM, which assumes no

[8] Controls include the binarized mean error, the Δ average switches, and the Δ average duration.

specific hypothesis on the regime structure. This particular relationship between KM and the regime matching will help understand further results.

Comparing (2) and (5), none of the models' performance is improved by matching the parameters, especially ST, which can often generate very unreasonable estimates, as noted in Section B.4. In contrast, matching the metrics explains a large part of the models' fixed effects, especially for ST, but not for KM. This is an important realization, since at first glance the effects of parameters and metrics could seem interchangeable, but they are not.

Table 7.1: Models' fixed effects - baseline and controls

	RMSE					
	(1)	(2)	(3)	(4)	(5)	(6)
MS	1.781***	1.226***	1.184***	1.492***	1.254***	1.244***
	(0.009)	(0.005)	(0.005)	(0.008)	(0.005)	(0.005)
SET	1.687***	1.271***	1.251***	1.327***	1.324***	1.250***
	(0.009)	(0.005)	(0.005)	(0.008)	(0.005)	(0.004)
ST	1.632***	1.219***	1.198***	1.370***	1.286***	1.197***
	(0.009)	(0.005)	(0.005)	(0.009)	(0.005)	(0.005)
KM	1.308***	1.068***	1.047***	0.984***	1.096***	1.067***
	(0.009)	(0.004)	(0.004)	(0.007)	(0.004)	(0.004)
R^2		yes		yes	yes	yes
regimes		yes	yes		yes	yes
Δ params.		yes	yes	yes		yes
Δ metrics		yes	yes	yes	yes	
Observations	79,934	62,013	62,013	68,124	62,016	71,054
R^2	0.614	0.886	0.884	0.795	0.880	0.876
Adjusted R^2	0.614	0.886	0.884	0.795	0.880	0.876
Residual Std. Error	1.277	0.466	0.469	0.878	0.477	0.492

Note: *p<0.1; **p<0.05; ***p<0.01

To understand how these fixed effects vary across different DGPs, Table 7.2 shows the fixed effects with no controls, but with stratifications in the dataset. It excludes the no-RS model. Column (1) is the same as before, but now includes the RF model, that is second to KM, but partially because of its 'generic' overfitted hyperparametrization. While the MS model showed a poorer relation with regime matching, in column (2) we can see that its flexibility pays off, as it performs better in asymmetric regimes. The RF model shows a similar behavior. In Column (3), ST deal better with small regime natures than SET. This is related to Table 6.1, where SET series were not fully separated in small regime natures, while in the ST ones they were, and it is reflected in the fixed effects.

Changes in μ (4) generate a better setup for separating the series via threshold, improving SET and ST. Changes in ρ (5) are poorly captured by ST, but nicely captured by MS and KM. In a σ change (6), everyone is worse, as it is overall harder to forecast. The overfitting of RF shows here, as it presents the highest change from column (1), while ST shows one of the best robustness, in line with its hability to deal with explosive volatility, as noted by Verne (2021).

The KM model's FE changes the least across stratifications, on average 0.17, followed by ST, with 0.23. These are characteristics of more general models, that have some 'flavour' of universal approximation power. The ability of ST to specially deal with some of the harder DGPs is related to its relationship to the complexity of a Neural Network, as discussed by Medeiros (2000), granted that the present setup with $S = 2$ does not its full potencial. The RF model is also kept shy of its potential, and while it would be interesting to also have a better architected RF, this benchmark already shows its improvement over the traditional models, while making it clear that overfit is a risk.

Table 7.2: Models fixed effects - across stratifications

	RMSE					
	(1)	(2)	(3)	(4)	(5)	(6)
MS	1.781***	1.308***	1.676***	1.782***	1.569***	1.987***
	(0.009)	(0.005)	(0.013)	(0.015)	(0.015)	(0.017)
SET	1.687***	1.322***	1.607***	1.601***	1.529***	1.932***
	(0.009)	(0.005)	(0.013)	(0.015)	(0.015)	(0.017)
ST	1.632***	1.313***	1.553***	1.391***	1.611***	1.884***
	(0.009)	(0.005)	(0.013)	(0.016)	(0.015)	(0.018)
KM	1.308***	1.132***	1.254***	1.182***	1.180***	1.562***
	(0.009)	(0.005)	(0.013)	(0.015)	(0.015)	(0.017)
RF	1.588***	1.214***	1.578***	1.320***	1.508***	1.910***
	(0.009)	(0.005)	(0.013)	(0.016)	(0.015)	(0.017)
Strat.	none	asym. RGP	small RN	μ RN	ρ_1 RN	σ RN
Observations	99,944	44,491	50,172	32,357	33,925	33,662
R^2	0.603	0.879	0.591	0.589	0.595	0.639
Adjusted R^2	0.603	0.879	0.591	0.589	0.595	0.639
Residual Std. Error	1.302	0.467	1.280	1.229	1.224	1.398
Note:						*p<0.1; **p<0.05; ***p<0.01

7.2 Model and DGP interaction

7.2.1 Mis-specification of RGP family

To study the effect of mis-specification, Table 7.3 defines mis-specification as "the family of the model being different than the family of the RGP". Only the MS, SET, and ST models are kept. The baseline effect of miss-specification is an RMSE increase of 0.515. To further understand how this effect changes across stratifications, consider the columns (2-6). (2): MS models suffer more, in line with previous results, while ST models suffer less. (3): Mis-estimating a RS model in a non-RS RGP is disastrous, and in light of that, mis-specification across RS RGPs is not so relevant. This will be further explored in the next subsection.

In (4), asymmetric regimes are harder to estimate, so that sometimes mis-specification helps. (5) Similar to before, σ changes make forecasting harder, and this effect is compounded by mis-specification; for the other two parameters the effect is similar. Finally, (6) shows that in small RNs, estimating the wrong regime generates a smaller error, so mis-specifying the RGP is not too problematic.

7.2.2 Specific model-RGP mis-specifications

Misspecification in general is bad. However, stratifying by each of the RGP families in Table 7.4 shows that each model (line) has similar values across the different RGPs (columns). This confirms that correctly specifying the RGP family is not the most relevant quality for performance.

Table 7.3: Models and DGPs mis-specification

	RMSE					
	(1)	(2)	(3)	(4)	(5)	(6)
Baseline	0.515^{***}					
	(0.013)					
Model: MS		0.627^{***}				
		(0.016)				
Model: SET		0.488^{***}				
		(0.016)				
Model: ST		0.420^{***}				
		(0.016)				
RGP: No RS			3.531^{***}			
			(0.014)			
RGP: MS			-0.016			
			(0.011)			
RGP: SET			0.006			
			(0.011)			
RGP: ST			-0.002			
			(0.011)			
RGP: asym.				-0.028^{***}		
				(0.006)		
RGP: sym.				0.020^{***}		
				(0.006)		
RN: μ					0.400^{***}	
					(0.016)	
RN: ρ_1					0.381^{***}	
					(0.016)	
RN: σ					0.760^{***}	
					(0.016)	
RN: small						0.424^{***}
						(0.015)
RN: big						0.607^{***}
						(0.015)
Constant	1.340^{***}	1.340^{***}	1.340^{***}	1.340^{***}	1.340^{***}	1.340^{***}
	(0.011)	(0.011)	(0.007)	(0.004)	(0.011)	(0.011)
Observations	59,092	59,092	59,092	53,009	59,092	59,092
R^2	0.027	0.029	0.554	0.001	0.037	0.029
Adjusted R^2	0.027	0.029	0.554	0.001	0.037	0.029
Residual Std. Error	1.424	1.422	0.964	0.538	1.417	1.422
F Statistic p-value	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001

Note: $^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$

With SET, the values are positive, meaning that the model does slightly well with no-RS series. For ST, the values are highly negative, meaning that no-RS series generate disastrous results, and this is carrying the result seen in Table 7.3 column (3). We can see that the oposite is true for KM, which actually does hugely well when there is no regime switching. For MS, RS or no-RS matters less. The invariance to RGP, specially between SET and ST is interesting, and in some ways goes agains the results of Aydın; Mermi (2022), although a bigger diversity of SET and ST models would be required for a proper comparison.

Table 7.4: Models and RGP pairs mis-specification

	RGP		
Model	MS	SET	ST
MS	0.001 (0.029)	0.005 (0.029)	0.007 (0.029)
SET	0.066** (0.03)	0.065** (0.03)	0.066** (0.03)
ST	-1.657*** (0.035)	-1.644*** (0.035)	-1.633*** (0.035)
KM	2.113*** (0.028)	2.05*** (0.028)	2.142*** (0.028)

Observations	100840
R^2, Adjusted R^2	0.826, 0.826
Resid. SE, F stat. p-value	0.884, <0.001

Note: *p<0.1; **p<0.05; ***p<0.01

RGP alone does not explain the mis-specification effect, but the interaction with RN was deemed as relevant in the exploratory analysis. Table 7.5 shows the coefficients for the interactions between model, RGP, and RN,[9] but we still have a similar result of 'invariance' within each 'panel' (RGP-model pair) of the table, except for KM.

7.2.3 Models and RC metrics

Fortunately, the point of this work is that the RC metrics are another way to characterize the DGP, and in some ways, a more general one than the RGP family 'label'. I construct similar tables interacting each model with the profiles of (dispersion of) metrics in the series. As the whole profile of metrics is important, all of the interactions of mod $\cdot\, d(\mu(.)) \cdot d(\rho_1(.)) \cdot d(\sigma(.))$ are included as controls.

Table 7.6 uses the true values of the metrics. After controling for the interactions, SET, ST and KM have a negative effect when estimated on series with high average separation, given the easier time to separate series via thresholds. For MS, we see don't have any significant effects, and not much can be said about it without a deepr analysis. ST seems to suffer from high separation in ACF. KM is also benefited by SD differenciation.

It is important to check if the estimated metrics generate similar results, as they are observable by the econometrician. That This is done in Table 7.7. Unfortunately, the results vary widely. As I have noted in Figure 6.4, the models do not perfectly match the metrics, which blurs the relationship found in the previous table.

Also recall, the metric estimation error varied across the unobservable RN, which means that these general relationships might not be too informative. Table 7.8 shows the values of interactions between models, metrics, and shich parameter changes.[10] The omitted group is the interaction with μ. We can

[9] Controls include are model, parameter change, and RGP family FEs, plus interactions between the last two.
[10] Controls include model, metric FEs, and interactions: metric^2 (with itself) and , model $\cdot$ metric^2.

Table 7.5: Models, RGPs, and RNs mis-specification

		Model			
RGP	SGP	MS	SET	ST	KM
No RS	μ	0.02 (0.048)	-0.055 (0.052)	2.085*** (0.091)	-2.868*** (0.047)
	ρ_1	-0.015 (0.047)	-0.034 (0.048)	1.927*** (0.052)	-1.522*** (0.047)
	σ	-0.031 (0.048)	0.048 (0.05)	1.825*** (0.056)	
MS	μ	-0.012 (0.038)	-0.008 (0.038)	0.012 (0.038)	-0.028 (0.038)
	ρ_1	-0.011 (0.038)	-0.004 (0.038)	-0.007 (0.038)	0.073* (0.038)
	σ	-0.003 (0.038)	-0.001 (0.038)	-0.004 (0.038)	
SET	μ	0.001 (0.038)	-0.01 (0.038)	0.043 (0.038)	-0.16*** (0.038)
	ρ_1	-0.011 (0.038)	-0.011 (0.038)	0.008 (0.038)	0.003 (0.038)
	σ	-0.003 (0.038)	0.006 (0.038)	-0.017 (0.038)	
ST	μ	-0.01 (0.038)	0 (0.038)	0.082** (0.038)	0.175*** (0.038)
	ρ_1	-0.006 (0.038)	-0.013 (0.038)	-0.013 (0.038)	-0.021 (0.038)
	σ	NANA (NA)	NANA (NA)	NANA (NA)	
Observations		100840			
R^2, Adjusted R^2		0.84, 0.84			
Resid. SE, F stat. p-value		0.848, <0.001			

Note: *p<0.1; **p<0.05; ***p<0.01

Table 7.6: Models and true RC metrics

	Metric		
Model	$d(\hat{\mu}(.))$	$d(\hat{\rho}_1(.))$	$d(\hat{\sigma}(.))$
MS	0.01 (0.034)	0.027 (0.344)	-0.012 (0.058)
SET	-0.108*** (0.034)	-0.163 (0.343)	-0.069 (0.058)
ST	-0.168*** (0.034)	0.911*** (0.349)	-0.006 (0.059)
KM	-0.183*** (0.033)	-0.155 (0.34)	-0.128** (0.057)
Observations	100840		
R^2, Adjusted R^2	0.613, 0.613		
Resid. SE, F stat. p-value	1.317, <0.001		

Note: *p<0.1; **p<0.05; ***p<0.01

Table 7.7: Models and estimated RC metrics

	Metric		
Model	$d(\hat{\mu}(.))$	$d(\hat{\rho}_1(.))$	$d(\hat{\sigma}(.))$
MS	-0.005*** (0.001)	0.213*** (0.02)	0.11 (0.186)
SET	-0.405 (0.275)	-0.104*** (0.018)	0.148 (0.186)
ST	0.363*** (0.023)	0.181 (0.186)	-0.41*** (0.032)
KM	0.041** (0.019)	0.14 (0.186)	-0.495*** (0.032)
Observations	88126		
R^2, Adjusted R^2	0.296, 0.296		
Resid. SE, F stat. p-value	1.017, <0.001		

Note: *p<0.1; **p<0.05; ***p<0.01

see, for example, that a big ACF difference is only a bad indicator when the parameter changing was σ, not ρ_1 (for SET and KM); average separation is often bad when μ is actually fixed.

Table 7.8: Models and estimated RC metrics - RN interactions

		Model			
Metric	SGP	MS	SET	ST	KM
$d(\hat{\mu}(.))$	ρ_1	0.025* (0.013)	-0.039*** (0.012)	0.107*** (0.023)	0.011 (0.009)
	σ	0.053*** (0.011)	0.025** (0.012)	0.113*** (0.023)	0.05*** (0.009)
$d(\hat{\rho}_1(.))$	ρ_1	-0.005 (0.008)	-0.01 (0.006)	-0.014* (0.007)	0.004 (0.015)
	σ	0.004 (0.008)	0.05*** (0.006)	0.013 (0.008)	0.119*** (0.016)
$d(\hat{\sigma}(.))$	ρ_1	-0.036*** (0.006)	0.053*** (0.011)	0.007 (0.015)	-0.006 (0.014)
	σ	-0.017*** (0.006)	0.061*** (0.01)	0.012 (0.014)	0.015 (0.013)
Observations		88126			
R^2, Adjusted R^2		0.737, 0.737			
Resid. SE, F stat. p-value		1.012, <0.001			

Note: *p<0.1; **p<0.05; ***p<0.01

The results of this section could be used as a practical recommendation to the econometrician, e.g. "if your KM-based estimated regimes have high volatility separation, the model selection is sound, as the others often struggle with such series". But this is currently very limited, because: (i) the metrics' estimation error hides the true relationships between model and metrics; (ii) as noted in Section 6.1, the three metrics used are not enought to fully characterize each DGP, thus the results vary widely across the unobservable RN; (iii) while the use of RC metrics abstract away from specific DGPs, the results are still conditional on the limited population used in this work.

Some strategies could be employed to improve the estimated metrics' meaning, such using weights proportional to how close each datapoint is to a regime change, considering other dispersion measures, even silhouette-like scores instead of simple absolute differences. On the front of characterizing the DGPs, more metrics should be added, such as 3rd and 4th moments, more ACF lags, more general distribution distance measures. Perhaps regime-unconditional and residual metrics – merging the usual model diagnostics in this analysis – could also improve the characterization. Overall the present exercise, while not perfect, is a step towards a more general and informative way to characterize the DGPs, and hopefully generate practical results for the econometrician.

7.3 Identification and performance

In the Models' Fixed Effect section, we talked about the qualities of the models. It can be interesting to analyze separately how matching each of the characteristics of the DGPs relates to the RMSE of an estimation. This is done in Table 7.9. Recall the definition of Δ done in the beginning of this section: a positive coefficient means that higher difference between the true and estimated is associated with a higher RMSE.

There are counterintuitive relationships between R^2, $BME(r)$ and RMSE: better fit, more error. These will be better analyzed below. The second column shows that ρ_1 can have an adverse effect, as it did infact had the least straightforward relationship with RGP, RN, and model interactions. A similar result is seen in the matching of metrics.

Table 7.9: RMSE and identification

	RMSE		
	(1)	(2)	(3)
R^2	0.411*** (0.003)	0.441*** (0.003)	0.435*** (0.003)
$BME(r)$	−0.098*** (0.003)	−0.089*** (0.003)	−0.092*** (0.003)
ΔSwitches	0.002 (0.003)	0.014*** (0.003)	−0.002 (0.003)
ΔDuration	−0.018*** (0.001)	−0.023*** (0.001)	−0.029*** (0.001)
$\Delta\mu$		0.029*** (0.003)	
$\Delta\rho_1$		−0.003** (0.002)	
$\Delta\sigma$		0.109*** (0.001)	
$\Delta d(\widehat{\mu}(.))$			0.023*** (0.003)
$\Delta d(\widehat{\rho}_1(.))$			−0.060*** (0.002)
$\Delta d(\widehat{\sigma}(.))$			0.141*** (0.002)
Constant	1.507*** (0.003)	1.406*** (0.003)	1.473*** (0.004)
Observations	92,001	91,960	82,922
R^2	0.231	0.282	0.286
Adjusted R^2	0.231	0.282	0.286
Residual Std. Error	0.780	0.753	0.768
F Statistic p-value	<0.001	<0.001	<0.001
Note:			*p<0.1; **p<0.05; ***p<0.01

To properly analyze the fisrt column, Table 7.10 stratifies the fit and regime effects by model. We can see that regime erros only have a negative relationship with no-RS and KM, two models that have some 'disregard' for the regime structure, while for the traditional models, smaller errors improve the performance. This is in line with Dacco; Satchell (1999), that found that minor errors in the regime variable can lead to large forecasting errors. Apparently, the KM is an exception to that rule, given the lack of parametric assumptions on the RGP. For the fit, the positive relationship still stands, this means that in line of fitting the regime, fitting the series is not a good guide of performance. The RF has the highest R^2 coefficient, another indication of its overfitting. MS has one similar to no-RS, atesting its flexibility, but not as small as KM.

Table 7.10: RMSE and R^2 across models

	RMSE	
	(1)	(2)
$R^2 \cdot$ No RS	0.749*** (0.009)	
$R^2 \cdot$ MS	0.746*** (0.009)	
$R^2 \cdot$ SET	0.851*** (0.010)	
$R^2 \cdot$ ST	0.849*** (0.013)	
$R^2 \cdot$ KM	0.467*** (0.009)	
$R^2 \cdot$ RF	1.482*** (0.011)	
$BME(r) \cdot$ No RS		−0.914*** (0.012)
$BME(r) \cdot$ MS		0.497*** (0.021)
$BME(r) \cdot$ SET		0.329*** (0.018)
$BME(r) \cdot$ ST		0.347*** (0.012)
$BME(r) \cdot$ KM		−0.310*** (0.016)
Observations	120,850	100,840
R^2	0.288	0.072
Adjusted R^2	0.288	0.072
Residual Std. Error	1.786	2.039
F Statistic p-value	<0.001	<0.001
Note:		*p<0.1; **p<0.05; ***p<0.01

To further analyze the effect of matching the metrics Table 7.11 shows model-metric interactions. MS has a massive improvement when matching the average, decent with volatility, but small with ACF. SET has a similar result, but overall matching is less important. ST has the smallest relationship with the average, but is big in ACF and the highest in volatility. KM is overall the most balanced, with a decent relationship with all metrics.

Table 7.11: RMSE and metrics identification across models

	Metric		
Model	$d(\hat{\mu}(.))$	$d(\hat{\rho}_1(.))$	$d(\hat{\sigma}(.))$
SET	0.737*** (0.02)	0.445*** (0.006)	0.359*** (0.008)
ST	0.36*** (0.007)	0.14*** (0.003)	0.421*** (0.007)
MS	0.228*** (0.005)	0.091*** (0.008)	0.428*** (0.008)
Observations	89065		
R^2, Adjusted R^2	0.441, 0.441		
Resid. SE, F stat. p-value	1.529, <0.001		

Note: *p<0.1; **p<0.05; ***p<0.01

Again, it might be important to check the interaction with RN, this is done in Table 7.12. The same controls are used. The effects are generally different when μ change, but not so much across ρ_1 or σ. Additionally, the interactions are overall smaller, indicating a smaller dependence on these non-observable characteristics of the DGPs.

These results could also be taken as practical recommendations, e.g. "if the econometrician has a belief about the true volatility dispersion, and the ST estimated doesn't match it, it might be a

Table 7.12: Models and estimated RC metrics - RN interactions

		Model			
Metric	SGP	MS	SET	ST	KM
$d(\hat{\mu}(.))$	ρ_1	-0.067*** (0.013)	0.012 (0.011)	0.127*** (0.021)	-0.004 (0.009)
	σ	-0.008 (0.011)	0.099*** (0.011)	0.154*** (0.021)	0.046*** (0.008)
$d(\hat{\rho}_1(.))$	ρ_1	-0.006 (0.014)	-0.025** (0.011)	-0.182*** (0.014)	-0.039 (0.026)
	σ	0.002 (0.013)	0.076*** (0.011)	-0.093*** (0.015)	0.073** (0.034)
$d(\hat{\sigma}(.))$	ρ_1	-0.048*** (0.01)	-0.04*** (0.015)	-0.074*** (0.025)	0.007 (0.027)
	σ	-0.028*** (0.009)	-0.074*** (0.015)	-0.143*** (0.025)	0.028 (0.024)
Observations		89065			
R^2, Adjusted R^2		0.736, 0.736			
Resid. SE, F stat. p-value		1.051, <0.001			

Note: *p<0.1; **p<0.05; ***p<0.01

problem". A substitute for belief would be a more general measure of variation of the metric across time, such as the SD of the rolling metric.[11] This interpretation is less usefull since such beliefs are not common, and we have the same external validity issues as before.

There are other match-able characteristics of the models, such as the τ for threshold models, the γ for ST, and transition probabilities for MS. They are not included here, as they are not directly comparable across models and do not generate interesting analysis.

7.4 Incorrect number of regimes

This section analyzes how the models perform when the assumed number of regimes $\hat{S}$ is different than the true one S. In this analysis, underestimating the number of regimes means estimating a simple $AR(1)$ with one of the two-regime RGPs, while superestimating means estimating some of the two-regime models with a series generator from a simple AR(1). The results should be interpreted as such.

In Table 7.13, the overall effect of an underestimation shows even a slight improvement, but, as I will show, this is carried by the fact that KM does super well when we have a no RS series.[^km-nors] With the superestimation, we have a positive result of 0.7.[12]

Column (2) has all of the metrics' interactions are negative. This means that when we superestimate, if the regimes were highly different in terms of the metrics, we have an attenuation of the bad effect of the misspecification. That is, it is very useful to be able to match these big differences, even if we need to use more regimes than could be needed.

When we consider the opposite scenario in (3), the interactions are still negative. But again, this is carried by the KM, as if we remove it from the sample, we get positive results throughout all the metrics. Column (4) shows that with underestimation, but similar regimes, we have an attenuation of the misspecification effect.

[11] The prediction power of this measure, for the true metric dispersion, must be studied.

[12] Controls include metrics FEs, and interactions of the S comparison with models, plus metrics with models, RGP family, and RN parameter.

Table 7.13: Mis specification of the number of regimes

	RMSE			
	(1)	(2)	(3)	(4)
$\hat{S} < S$	-0.809^{***} (0.017)		-0.350^{***} (0.021)	-0.494^{***} (0.023)
$\hat{S} > S$	0.714^{***} (0.020)	1.981^{***} (0.028)		
$\hat{S} > S : d(\hat{\mu}(.))$		-1.927^{***} (0.029)		
$\hat{S} > S : d(\hat{\rho}_1(.))$		-2.166^{***} (0.096)		
$\hat{S} > S : d(\hat{\sigma}(.))$		-2.727^{***} (0.050)		
$\hat{S} < S : d(\hat{\mu}(.))$			-0.520^{***} (0.028)	0.301^{***} (0.032)
$\hat{S} < S : d(\hat{\rho}_1(.))$			0.002 (0.097)	0.426^{***} (0.111)
$\hat{S} < S : d(\hat{\sigma}(.))$			-0.343^{***} (0.047)	0.517^{***} (0.054)
Constant	1.479^{***} (0.013)	1.183^{***} (0.013)	1.446^{***} (0.014)	1.580^{***} (0.017)
Observations	81,970	81,970	81,970	61,128
R^2	0.498	0.519	0.492	0.537
Adjusted R^2	0.498	0.519	0.492	0.537
Residual Std. Error	0.938	0.918	0.943	1.005
F Statistic p-value	<0.001	<0.001	<0.001	<0.001
Note:				*p<0.1; **p<0.05; ***p<0.01

To stratify the effects by model, consider Table 7.14. As noted before, the ST model deals very bad with single-regime data, while the KM actually benefits from it. The MS is a little more robust than SET. Given the specific definition of overestimation, this result is not perfectly comparable with Janczura; Weron (2010), thar found a percieved big sensitivity of $\hat{S}$ in the MS model.

Table 7.14: Mis specification of the number of regimes

	RMSE
$\hat{S} > S$: MS	1.152^{***} (0.022)
$\hat{S} > S$: SET	1.177^{***} (0.024)
$\hat{S} > S$: ST	3.072^{***} (0.031)
$\hat{S} < S$: KM	-0.964^{***} (0.022)
Constant	1.297^{***} (0.011)
Observations	100,840
R^2	0.564
Adjusted R^2	0.564
Residual Std. Error	0.888
F Statistic p-value	<0.001
Note:	*p<0.1; **p<0.05; ***p<0.01

8 Conclusion

This work studied regime switching models and DGPs from a learning perspective, showing how the regimes' distributions can be used to analyze them, and generate stilyzed facts relevant to forecast performance.

The general and expandable framework plus implementation is a contribution in itself. I represent any RS DGP as a combination of a regime generating process (RGP) and a series generating process (SGP), and I formalize regime-conditional (RC) metrics as functions that characterize the regime distributions. This makes it possible to discuss model behavior both in terms of labels (RGP and regime nature) and in terms of observable regime characteristics computed from $(\hat{y},\ \hat{r})$.

The framework was implemented with a constrained subset of its full capacity. I restrict attention to stationary Gaussian $AR(1)$ SGPs, two regimes, a MS, SET and ST of RGPs, and a minimal set of RC metrics (the first two moments and the lag-1 autocorrelation), summarized via distance across regimes.

Some facts about the DGPs' regime separation were presented. For the MS RGP, the metrics define a clear-cut profile: in μ changes only the average separates the regimes, in ρ_1 changes it is both the ACF and SD, and in σ changes, only the SD. For SET and ST, the interaction between the RGP and the RN creates more complex profiles, where all metrics are different across regimes, and more metrics would be required to fully identify the DGP. The assymetric RGPs have a similar separation, but require larger sample size to estabilish it. With approximately 60 observations, the separation of these stationary $AR(1)$ series with $\sigma = 1$ converged. The RMSE distributions of observations with correctly identified regimes have slimmer tails, but not always, especially for the MS model.

On the performance of the models, the MS model has a worse overall fixed effect, but its flexibility allows it to commit fewer egregious errors overall, performing better in asymmetric regimes and showing the smallest relationship between R^2 and RMSE. The SET and ST models have better fixed effects and fare better for intercept changes. ST is more robust to small parameter changes, but commits huge errors when estimated on a no-RS RGP.

The baseline effect of mis-specification is an RMSE increase of 0.547. The σ changes make forecasting harder, and this effect is compounded by mis-specification. In small RNs, mis-specifying the RGP is not too problematic. Mis-specification across RS RGPs is not so relevant, and no specific model-RGP pair has a different fixed effect than the correctly specified pair.

Interactions of model-RGP-RN are similarly insignificant, but the RC metrics are another way to characterize the DGP. SET and ST have a negative effect when estimated on series with high average separation. For MS, we have an improvement given average, but a worsening given ACF separation. The opposite is true for ST, but without statistical significance. Finally, high SD has insignificant effects, but negative for ST and MS. With the estimated metrics, the results are mostly the same. On top of the universal limitations, these general results are not invariant to non-observable DGP characteristics.

Several analyses can be done to expand the results. A main one is to understand how the effect of mis-specifying the number of regimes relates to the regime separation. Series with lower regime separation can often be estimated with fewer regimes without a large performance loss, generating a recommendation for this hyperparameter selection. Another is to include less-parametric models in the analysis, such as a Random Forest benchmark and a Clustering-based RS model.

The main limitation of the current implementation is external validity. Even when the analysis is phrased in terms of RC metrics, it remains conditional on a narrow population of DGPs and on a limited set of regime descriptors. A more robust assessment of the usefulness of RC metrics for the econometrician requires expanding both sides: richer DGPs (more SGP functional forms, more regimes, additional regime mechanisms, non-Gaussian errors, and possibly non-stationary settings) and richer metrics (higher moments, silhouette-like differences, distribution-distance or separation criteria). This is the natural next step to turn the RC-metric approach into a more general model-selection and diagnostic tool.

Some results are expected, but interesting light has been shed on (i) the relationship between RGPs and RNs, especially via the metrics separation discussion; (ii) the differences between MS and SET/ST; and (iii) the effects of mis-specification. The performance of models given regimes characteristics could be of high practical use for the econometrician, but must be taken with caution and requires further investigation.

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AI disclaimer: this work was generated generally without the help of large language models, except sparingly as a research tool and code autocompletions during the implementation phase.

A DGPs, models, and metrics

A.1 RGPs and models

A.1.1 No RS (noRS)

Hypothesis: No regime switching, always at regime '1'.

$$\begin{aligned} &r_t^1(.) = 1 \\ &r_t^s(.) = 0,\ s \in \{2, \dots, S\} \end{aligned} \tag{A.1}$$

Empirical model: a simple $AR(1)$ model, estimated via OLS, with no regime-switching component.

A.1.2 Structural break (SB)

Hypothesis: Regime changes at specific time points $\tau \in (1:T)^{S-1}$. Formally:

$$\begin{aligned} &r_t^s(.\ ;\ \tau) = \mathbb{1}(\tau'_{s-1} < t \leq \tau'_s), \quad \forall s \in \{1, \dots S\} \\ &\tau \in \mathbb{N}^{S-1}, \quad \tau_s > \tau_{s-1}\ \forall s, \quad \tau' = (0, \tau, T) \end{aligned} \tag{A.2}$$

Empirical model: Given τ, the model estimates μ and ρ_1 via OLS in each regime. τ is chosen by minimizing the sum of squared residuals over a grid search of breakpoints.

Similarly defined by Bai; Perron (1998). Review of other options by Casini; Perron (2018). This RGP and model was removed from the final analysis.

A.1.3 Self-exciting threshold (SET)

Hypothesis: Regime changes when the series, possibly at a lag $d \in \mathbb{N}^*$, crosses specific threshold values $\tau \in \mathbb{R}^{S-1}$. Transformations of the variable can be considered.[13] Formally:

$$\begin{aligned} &r_t^s(.\ ;\ (\tau, d, g)) = \mathbb{1}(\tau'_{s-1} < g(y)_{t-d} \leq \tau'_s), \quad \forall s \in \{1, \dots S\} \\ &\tau \in \mathbb{R}^{S-1}, \quad \tau_s > \tau_{s-1}\ \forall s, \quad \tau' = (-\infty, \tau, \infty), \quad d \in \mathbb{N}^* \end{aligned} \tag{A.3}$$

Empirical model: Given τ and d, the model estimates μ and ρ_1 via OLS in each regime. τ and d are chosen by minimizing the sum of squared residuals over a grid search of breakpoints and lags. One can also leave d fixed.

Similarly defined by Tong; Lim (1980). Review of other options by Chen; Liu; So (2011).

[13] For example, $g(x) = |x|$ or $g(x) = \Delta x$.

A.1.4 Smooth transition (ST)

Hypothesis: Regime changes smoothly, with a continuous function g, often a CDF, based on the difference between the series and the threshold $\tau \in \mathbb{R}$, possibly at a lag $d \in \mathbb{N}^*$. Medeiros (2000) has shown that a generalization to S regimes is a neural network, but currently, I only consider $S = 2$. Formally:

$$\begin{aligned} &r_t^1(. \,;\, (\tau, d, g)) = g(y_{t-d} - \tau), \quad r_t^2(. \,;\, (\tau, d, g)) = 1 - r_t^1(. \,;\, (\tau, d, g)) \\ &\tau \in \mathbb{R}, \quad d \in \mathbb{N}^* \end{aligned} \tag{A.4}$$

Often, the function g depends on a smoothness parameter γ, i.e., when $\gamma \to \infty$, $g \to \mathbb{1}$. This parameter can be jointly estimated with the others.

Empirical model: Estimated via non-linear squares of the residuals, over μ, ρ_1 (for each regime), τ, and γ. Uses some numerical optimization, which depends on starting values and does not guarantee a global optimum.

Similarly defined by Teräsvirta (1994). Review of other options by Dijk; Teräsvirta; Franses (2002)

A.1.5 Markov-Switching (MS)

Hypothesis: Regime changes stochastically, following a Markov process with transition matrix $\Gamma \in [0,1]^{S \times S}$. The probability of being in regime s at time t depends only on the regime at time $t-1$, often with Γ implying some persistence. Formally:

$$\begin{aligned} &r_t^s(. \,;\, \Gamma) \sim P(r_t^s = 1 | r_{t-1}) =: \Gamma_{s, r_{t-1}} \\ &\Gamma \in [0,1]^{S \times S}, \quad \textstyle\sum_{i=1}^{S} \Gamma_{s,i} = 1 \ \forall s \end{aligned} \tag{A.5}$$

Empirical model: There are multiple algorithms, including maximum likelihood estimation, expectation maximization, and Markov chain Monte Carlo methods. The EM algorithm uses Kalman to find smoothed probabilities of r, then the conditional probabilities given the current guess of parameters, then the guess of parameters is updated via maximizing the likelihood given the probabilities. These two steps are iterated until convergence.

Similarly defined by Hamilton (1989). Review of other options by Song; Woźniak (2021).

A.1.6 Unsupervisioned clustering (UC)

Hypothesis: there is RS, but no hypothesis is mabe about the RGP.

Model: Unsupervised clustering techniques, such as K-Means, can be used to estimate the regimes based on y_t, its lags, and rolling moments. Given the regimes, μ and ρ_1 are estimated via OLS. This hybrid approach yields non-standard asymptotic properties. Other clustering techniques could be used, but I focus on the general K-Means clustering problem is as below:

$$\begin{aligned} &\hat{r}_t^s(. \,;\, (\mathrm{norm}, \mathrm{centroid})) = \mathbb{1}(y_t \in R_s) \\ &R = \arg\min_{R'} \textstyle\sum_{s=1}^{\hat{S}} \sum_{y_t \in R'_s} \mathrm{norm}(y_t - \mathrm{centroid}(R'_s)) \end{aligned} \tag{A.6}$$

Similarly defined by Akioyamen; Tang; Hussien (2020). More clustering techniques reviewd by Paparrizos; Yang; Li (2024).

A.1.7 Random forests (RF)

Hypothesis: there is no RS, the non-linearity is captured by the tree and ensamble structure of the RF.

Model: a RF is estimated based on y_t, its lags, and rolling moments.

A review of the time series RF literature is done by Hu; Szymczak (2022).

A.2 Metrics

The estimated conditional mean and standard deviation can be calculated as, respectively:

$$\hat{\mu}(y, r|s) := \sum_{t=1}^{T} r_t^s \cdot y_t \tag{A.7}$$

$$\hat{\sigma}(y, r|s) := \sqrt{\frac{1}{1 - \sum_{t=1}^{T} (r_t^s)^2} \sum_{t=1}^{T} r_t^s \cdot (y_t - \hat{\mu}(y, r|s))^2} \tag{A.8}$$

Note the bias correction factor in the denominator of the RC SD, which is necessary given the estimated mean.

As noted, in the case of binary r_t, only the observations of regime s have non-zero weights, and the formulas are respectively equivalent to:

$$\frac{1}{|R_s|} \sum_{y_t \in R_s} y_t, \qquad \sqrt{\frac{1}{|R_s| - 1} \sum_{y_t \in R_s} (y_t - \hat{\mu}(y|s))^2}$$

For a regime-conditional moment of $(y_t,\ y_{t-j})$, we must define the notion of 'being in the same regime'. Consider $r_t^s \cdot r_{t-j}^s$, which has a correct 'truth table' for binary regimes, but also has an interpretation for continuous ones: when closer to 1, the higher the weight of both y_t and y_{t-j} being in regime s. But, this ignores that fact that y_t and y_{t-j} can be in the same regime, but in different regime instances. To account for that, the correct weighting should consider the whole window of $y_{t-j}, \cdots, y_t$:

$$\hat{\rho}_j(y, r|s) = \frac{\sum_{t=1+j}^{T} \left(\prod_{k=1}^{j} r_k^s\right) \cdot (y_t - \hat{\mu}(y, r|s)) \cdot (y_{t-j} - \hat{\mu}(y, r|s))}{\sum_{t=1}^{T} \left(\prod_{k=1}^{j} r_k^s\right) \cdot (y_{t-j} - \hat{\mu}(y, r|s))^2} \tag{A.9}$$

Note the absence of bias correction. While it could be present, it can generate larger-than-one correlations, and is often omitted.

For binary regimes, this is equivalent to calculating the unweighted autocorrelation of every concurrent window of regime s.

In this work, I am currently using the binary version of the RC metrics, calculated after binaryzing r.

Recall that a RC metric returns a sequence with entries for each regime, so when describing the e.g. RC mean $\mu(y, r)$, I am refering to:

$$\mu(y,r) := \mu(y,r|S) = (\mu(y,r|s))_{s\in 1:S} \tag{A.10}$$

A.2.1 True moments of the considered DGPs

Given the weakly stationary within regimes assumption, the regime-conditional moments are independent of the RGP, and are the simple $AR(1)$ moments:

$$\begin{aligned} \mu(y_t|s) &\equiv E[y_t|y_t \in R_s] = \tfrac{\mu^s}{1-\rho_1^s} \\ \sigma(y_t|s) &\equiv Var[y_t|y_t \in R_s] = \sqrt{\tfrac{(\sigma^s)^2}{1-(\rho_1^s)^2}} \\ \rho_j(y_t|s) &\equiv Corr[y_t, y_{t-1}|y_t \in R_s] = (\rho_1^s)^j, \quad j \in \mathbb{N}^* \end{aligned} \tag{A.11}$$

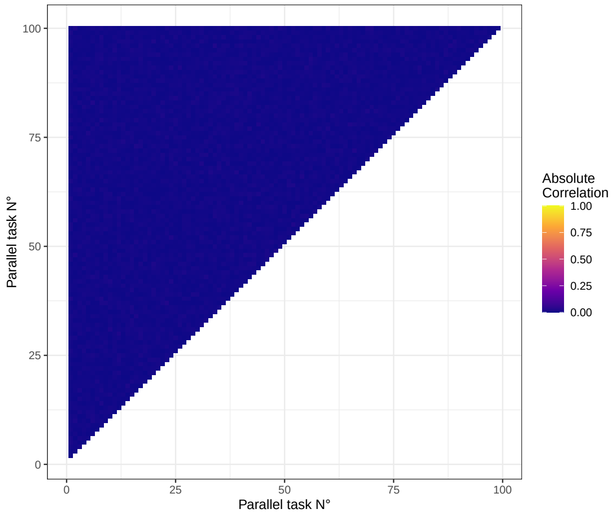
As described in Section 3.3.2, there can be better estimators for populational RC metrics than the ones defined in the above section. One can simply plug in the estimated parameters in the equation above to get a better estimator of the moments. In this work, I use this approach, which is also more computationally efficient.

B Diagnostics

B.1 Random errors

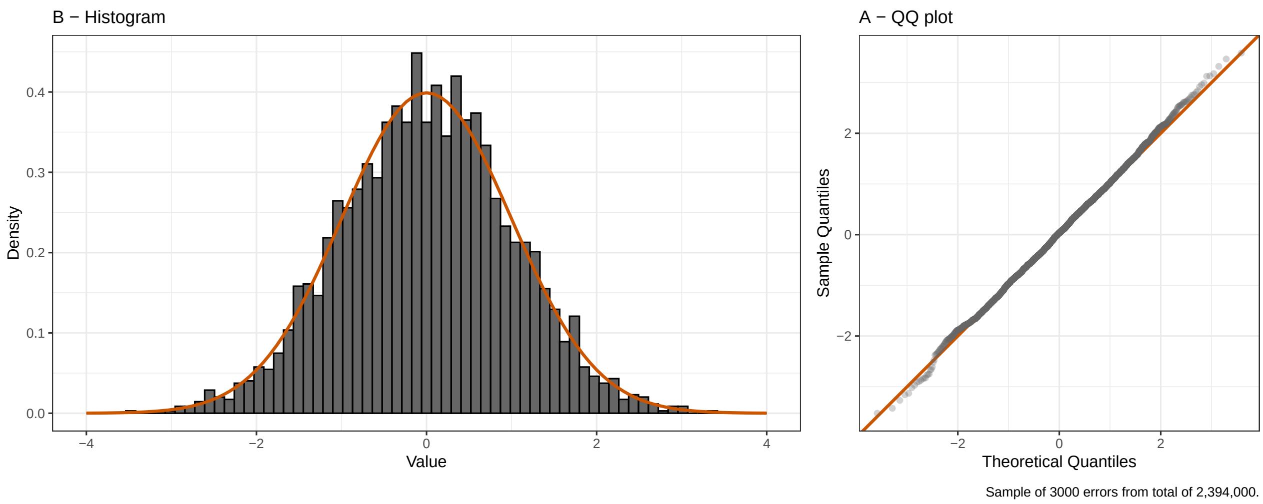
The Figure B.1 shows the correlation of the errors across the parallelization structure. A simple visual check shows no evident patterns and an overall low correlation, as expected.

Figure B.1: Random errors - Correlation across parallelization structure



The Figure B.2 shows the distribution of a size 3000 sample of the errors, via the usual histogram and QQ-plot. The distribution is very close to normal, as expected.

Figure B.2: Random errors - Distribution



B.2 Estimated errors

The Figure B.3 shows the distribution of the estimation errors (residuals) across models, while Figure B.4 shows the distribution of forecasting errors. Overall the distributions are as expected, and we can see that the former has fatter tails than the latter. In the former, approximately 10k observations are outside the range of the x-axis, some of which were considered outliers.

Figure B.3: Residuals - Distribution

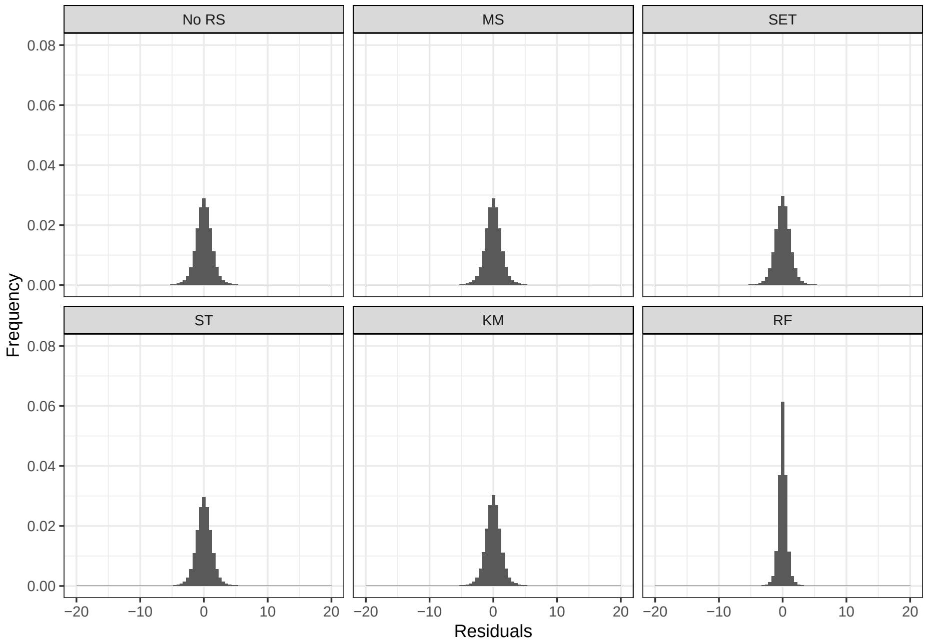
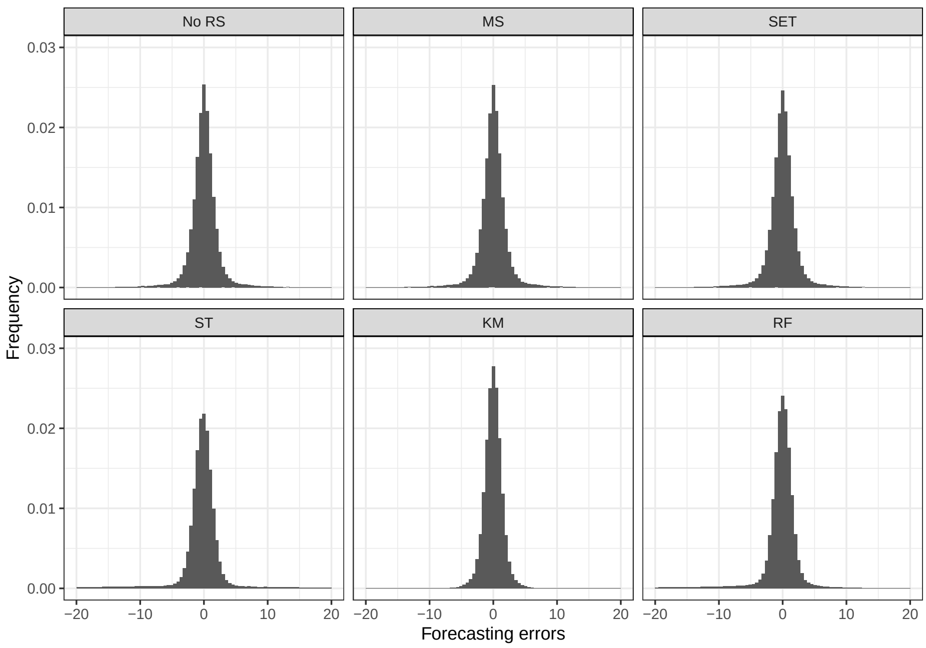


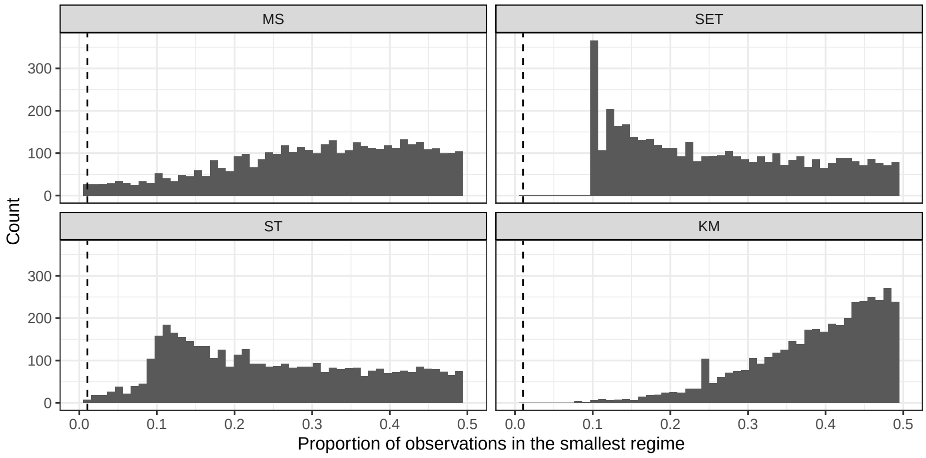
Figure B.4: Forecasting errors - Distribution



B.3 Regime proportions

The Figure B.5 shows the distribution of the proportion of the lowest frequent estimated regime, separated per model. Estimations below the dashed line had only two observations in the regime and had to be removed.

Figure B.5: Regime proportion - Distribution



B.4 Parameters and model metadata

Figure B.6 shows the distribution of the estimated parameters across models and parameter. All the by-regime values of a parameter are clumped together in the same panel. Overall, the distributions are as expected, with approximately 10.000 values being outside the range of the x-axis, some of which were considered outliers.

Figure B.7 shows the distribution of RGP-related metadata, such as the MS transition probabilities, ST γ, and SET τ. Overall, the distributions are as expected.

Table B.1 compares the models parameters to the true values of the DGPs. Each group of lines corresponds to the moments of a DGP. The first two columns relate to the values conditional on regime 1 and 2, the third column gives the unconditional values. Each cell has the value of the moment, and in brackets the p-value of the null hypothesis that the moment is equal to its true value. The table uses only symmetric RGPs and big SGPs. Note that the moments don't need to be exactly the same, since all the models allow for all the parameters to change, a different assumption than that of the regime natures.

B.5 Metrics calculation

The calculated metrics can be seen in Table B.2. Each group of lines corresponds to the moments of a DGP. The first two columns relate to the values conditional on regime 1 and 2, the third column gives the unconditional values. Each cell has the value of the moment, and in brackets the p-value of the null hypothesis that the moment is equal to its true value. The table uses only symmetric RGPs and big SGPs. Note that the moments don't need to be exactly the same, since all the models allow for all the parameters to change, a different assumption than that of the regime natures.

Figure B.6: Parameters - Distribution

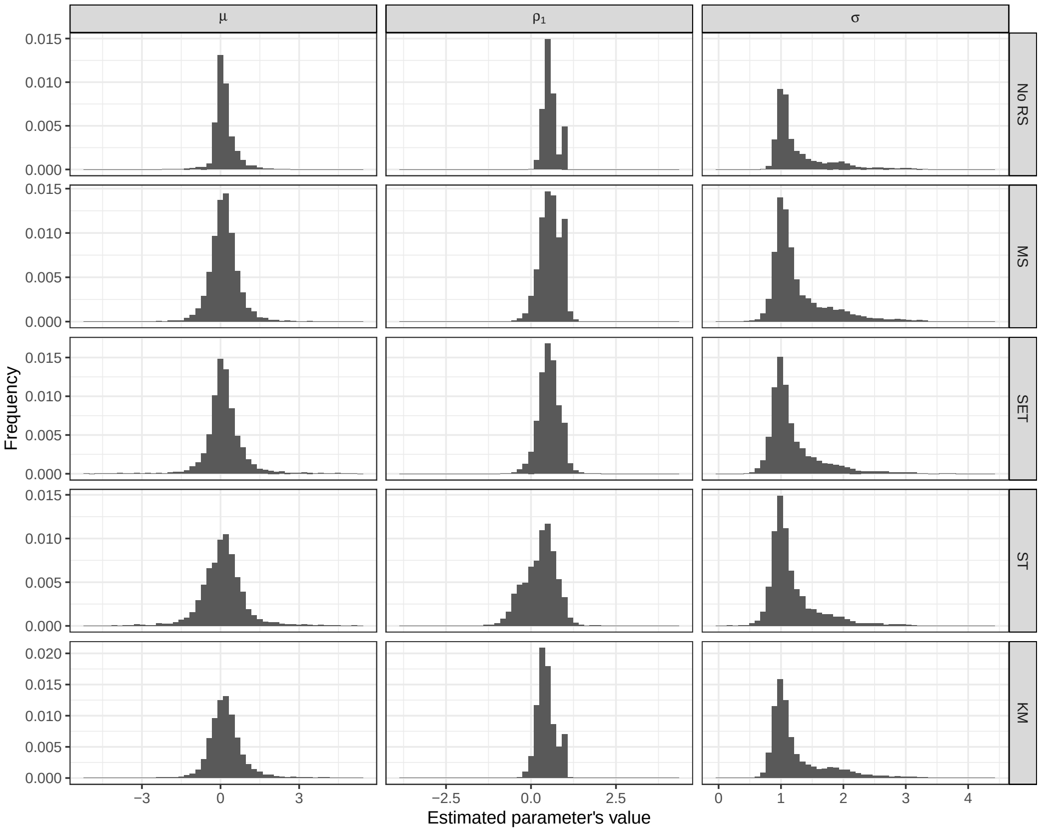


Figure B.7: Model metadata - Distribution

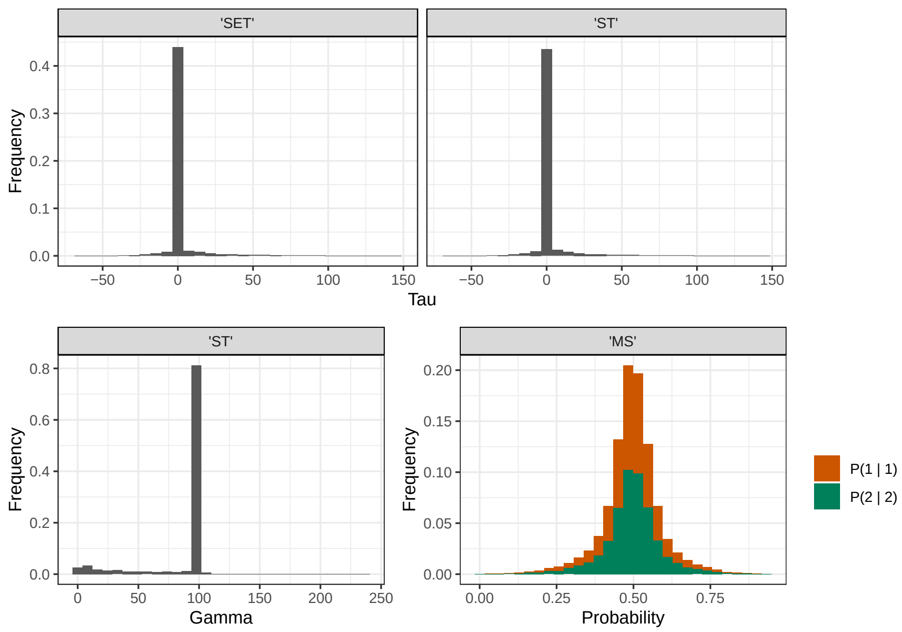


Table B.1: Estimated coefficients across DGPs

		μ^s		ρ_1^s		σ^s	
RGP	RN	$s = 1$	$s = 2$	$s = 1$	$s = 2$	$s = 1$	$s = 2$
MS, $p_{21} = .1$	μ $(0, 1)$	.16*** (.011)	.6*** (.015)	.68*** (.008)	.57*** (.008)	1.07*** (.007)	1.07*** (.007)
	ρ_1 $(.4, .6)$	-.01 (.012)	.03** (.011)	.32*** (.011)	.74*** (.008)	1.04*** (.006)	1.04*** (.006)
	σ $(1, 2)$	.01 (.016)	-.01 (.017)	.46** (.014)	.49 (.014)	1.42*** (.011)	1.7*** (.012)
SET, $\tau = .5$	μ $(0, 1)$	.22*** (.018)	1.11*** (.028)	.74*** (.01)	.5 (.014)	.98*** (.006)	.97*** (.006)
	ρ_1 $(.4, .6)$	.29*** (.018)	.04* (.021)	.36*** (.01)	.77*** (.009)	.97*** (.006)	.96*** (.005)
	σ $(1, 2)$	-.05* (.023)	.02 (.024)	.47** (.014)	.5 (.013)	1.19*** (.008)	1.46*** (.009)
ST, $\tau = .5$	μ $(0, 1)$	-.19*** (.027)	.77*** (.017)	.17*** (.021)	.11*** (.02)	.97*** (.006)	.95*** (.006)
	ρ_1 $(.4, .6)$	-.13*** (.032)	.04 (.022)	-.35*** (.018)	.71*** (.01)	.97*** (.005)	.96*** (.006)
	σ $(1, 2)$	-.05 (.043)	-.02 (.044)	.26*** (.021)	.25*** (.021)	1.39*** (.009)	1.64*** (.009)

Note: *p<0.1; **p<0.05; ***p<0.01

Table B.2: Estimated metrics across DGPs

RGP	RN	$\hat{\mu}(.\|s)$ $s=1$	$\hat{\mu}(.\|s)$ $s=2$	$\hat{\mu}(.\|s)$ $\perp s$	$\hat{\rho}_1(.\|s)$ $s=1$	$\hat{\rho}_1(.\|s)$ $s=2$	$\hat{\rho}_1(.\|s)$ $\perp s$	$\hat{\sigma}(.\|s)$ $s=1$	$\hat{\sigma}(.\|s)$ $s=2$	$\hat{\sigma}(.\|s)$ $\perp s$
MS, $p_{21} = .1$	μ $(0, 1)$	.21*** (.012)	1.79*** (.011)	1.03 (.014)	.38*** (.005)	.38*** (.005)	.55 (.003)	1.18*** (.007)	1.18*** (.006)	1.41 (.006)
	ρ_1 $(.4, .6)$	0 (.008)	.05** (.022)	.03 (.012)	.16*** (.006)	.57*** (.004)	.51 (.004)	1.02 (.005)	1.49*** (.012)	1.31 (.008)
	σ $(1, 2)$	-.01 (.012)	0 (.021)	0 (.013)	.38*** (.005)	.38*** (.005)	.44 (.003)	1.2*** (.007)	2.22*** (.012)	1.81 (.01)
SET, $\tau = .5$	μ $(0, 1)$	-.19*** (.014)	2.07*** (.009)	1.47 (.016)	.2*** (.008)	.37*** (.004)	.6 (.002)	1.05*** (.007)	1.11*** (.005)	1.47 (.008)
	ρ_1 $(.4, .6)$	-.08*** (.007)	1.29*** (.014)	.63 (.013)	.08*** (.006)	.39*** (.006)	.54 (.003)	1*** (.005)	1.2*** (.007)	1.3 (.007)
	σ $(1, 2)$	-.43*** (.007)	.85*** (.02)	-.01 (.012)	.19*** (.005)	.17*** (.007)	.44 (.003)	1.1*** (.004)	2.06*** (.012)	1.59 (.009)
ST, $\tau = .5$	μ $(0, 1)$	1.02*** (.006)	.56*** (.008)	.85 (.006)	.12*** (.005)	.04*** (.007)	.27 (.004)	1.04*** (.004)	1.01*** (.005)	1.05 (.003)
	ρ_1 $(.4, .6)$	.41*** (.009)	-.66*** (.011)	-.39 (.011)	-.01*** (.008)	.45*** (.005)	.53 (.003)	1*** (.006)	1.23*** (.007)	1.27 (.007)
	σ $(1, 2)$	.81*** (.01)	-.59*** (.014)	0 (.013)	.1*** (.006)	.24*** (.005)	.44 (.003)	1.33*** (.006)	1.89*** (.009)	1.81 (.007)

Note: *p<0.1; **p<0.05; ***p<0.01